

A multimodal field dataset and semantic knowledge graph for the Rub' al-Khali desert microbiome

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Abstract

Environmental microbiome reuse requires field, molecular and laboratory data to remain connected to provenance. We describe a multimodal resource from five expeditions across a 1,043 km cumulative path (1,015 km end to end) in the Rub' al-Khali from March 2023 to October 2025. A 2,550-row source ledger connects specimens, DNA extractions, libraries, runs and quality-controlled community profiles; 2,516 non-control records are sample individuals in an SIO-based OWL

graph. The canonical feature table contains 1,271 profiles; 1,237 pass ecological filtering across 64 numeric site labels. Additional records comprise 274 environmental logs, 12,936 monthly climate measurements, 712 admitted archived-soil pH observations, and two uncalibrated X-ray fluorescence workflows (71 field sessions at 58 sites and 725 laboratory records). Shape Expressions, bounded consistency checks and an executable query validate stated structural coverage. The package supports provenance-aware reuse, not an operational digital twin.

Keywords: Knowledge Graph, Microbiome, Rub' al Khali, SIO, Semantic Web

1 Background and summary

The Rub' al-Khali (Empty Quarter) is a hyper-arid sand sea of approximately 500,000 km² across the southern Arabian Peninsula [1]. Air temperatures recorded during these expeditions span 20.0–49.4°C across 273 field observations, and the region receives negligible and highly irregular rainfall. Its microbial communities remain poorly characterized relative to other drylands [2]. To predict ecosystem responses to climate change and to inform conservation and land-use policy, researchers require deep knowledge of the biogeochemistry of such extreme environments [3].

Saudi Arabia already has a substantial arid-soil microbiome literature. Previous studies examined elevation gradients [4], rainfall responses across soil compartments [5], open and rhizosphere soils across three regions [6], rhizosphere bacteria of pioneer sand plants near Jizan [7], and bacteria associated with native desert plants [8]. More recent surveys characterized fertility islands and plant-associated networks [9, 10], sandy soils near natural and artificial water points [11], rhizosphere and endosphere communities across northern reserves [12], and Red Sea coastal sabkhas [13]. Fungal studies progressed from four western-coastal composite samples and four Saudi open-sand samples [14, 15] to a standardized single-visit survey of 85 locations across Saudi Arabia [16]. A recent synthesis likewise identifies substantial geographic and taxonomic gaps in Arabian biological-soil-crust research [17]. In the United Arab Emirates, related studies profiled a Rub' al-Khali interdune microbial mat [18] and desert-plant rhizosphere and endosphere communities [19]. This resource does not claim the first Saudi, national-scale or arid-soil survey. It connects a repeated five-expedition bacterial transect to joined provenance across locations, physical specimens, molecular products, environmental observations, climate, and distinct field and laboratory geochemistry workflows.

To characterize complex ecosystems, researchers must integrate heterogeneous and multi-scale data: microbial community profiles derived from DNA sequencing, elemental soil composition from geochemical analyses, spatiotemporal environmental variables, and provenance metadata that link physical samples to their digital representations. Traditional approaches store these data types in separate, disconnected repositories, which limits cross-domain queries and reduces reproducibility.

Community reporting and infrastructure efforts provide important precedents. The FAIR principles set the reuse target that a deposited resource is judged against [20].

MIxS/MIMARKS define contextual fields for marker-gene and environmental sequence data [21], and the 67-item STREAMS guideline extends reporting practice to environmental and non-human-associated microbiomes [22]. The National Microbiome Data Collaborative (NMDC) demonstrates how persistent identifiers, standardized metadata, linked-data schemas, and reproducible workflows can support reuse across environmental microbiome studies [23]. Three vocabularies cover parts of what an expedition resource must express: PROV-O models derivation and activity [24], OBI models investigations and assays [25], and SOSA/SSN models observations and their procedures [26]. None of them, on its own, carries a field specimen through extraction, library preparation, sequencing, quality control and taxon observation while keeping the geochemical and climate measurements attached to the same site individual. The residual gap after MIxS, STREAMS and NMDC is therefore not vocabulary coverage but the joined instantiation: a single graph in which every one of those records resolves to the same specimen and the same location.

Knowledge graphs address a complementary integration problem. KG-Microbe builds modular graphs that connect microbial taxa with traits, environments, and growth preferences [27]. MetagenomicKG integrates seven taxonomic, functional, pathogen, and biomedical sources for metagenomic applications [28]. These resources establish that the existence of a microbiome knowledge graph is not itself novel. The contribution of the present resource is instead the joined, expedition-level provenance of locations, physical specimens, laboratory processes, geochemistry, climate, sequence runs, quality control, and taxon observations from a single landscape survey.

Constructing that resource requires a unified representation of heterogeneous instruments and protocols, preservation of original taxonomic assertions alongside harmonized identifiers, structural validation at ingest, and an explicit account of which entailments and queries are actually supported. We use persistent nodes for samples, processes, qualities, quantities, and digital records so that data users can traverse the recorded provenance chain instead of reconstructing it from unrelated tables. The chain is documented where the source identifiers permit it: the field XRF log carries no physical specimen identifier, and the duplicated export columns described in Section 3 have no recorded upstream relationship, so those links are absent rather than inferred.

The dataset covers five field expeditions (2023–2025) along a transect of more than 1,000 km across the Saudi Rub’ al-Khali. The intended repeated frame comprises numeric sites 1–60; four additional numeric identifiers were used only on Trip 1, and the site catalogue contains named special-interest locations. Coordinate matching, confirmed against the field records, establishes that numeric identifiers 61–64 denote four of those named catalogue locations. We retain the numeric identifiers as source provenance but do not count them as four additional physical locations. The current source ledger contains 2,550 rows: 2,302 master-sheet rows (2,268 non-control and 34 control rows) plus 248 plant rows. These are source records rather than a claim that 2,550 distinct specimens have entered every downstream product. Sequence records, quality-controlled community profiles, and derived taxonomic observations are distinguished from physical specimens throughout this descriptor. The generated graph contains 2,516 sample individuals. Ecological filtering retains 1,237 profiles across 64

numeric site labels, four of which are Trip-1-only aliases of named catalogue locations rather than additional physical sites: 1,227 from the primary repeated-campaign frame of sites 1–60 and 10 Trip-1-only profiles from sites 61–64. Site measurements and Open-Meteo climate records [29] are linked at the site-expedition level.

The geochemistry comprises two complementary X-ray fluorescence (XRF) workflows. During Trip 5, the field log contains 106 in-situ handheld entries across 59 sites; 71 complete sessions at 58 sites were retained, including nine sites with repeated complete sessions. The log has no physical-sample identifier. In the laboratory, archived material from all five trips was represented by 725 processed records: 547 from Trips 1–4 and 178 from Trip 5. All 725 enter the canonical laboratory analysis. A former 158-row Trip 5 table and the resulting 705-row cross-trip table are retired legacy artifacts and are not analytical inputs.

The knowledge graph uses the SemanticScience Integrated Ontology (SIO) [30] as its upper-level framework. Domain-specific concepts are represented as subclasses of SIO entities, and measurements follow SIO’s process–quantity–quality pattern. Chemical entities are grounded where supported in ChEBI [31], environmental context in ENVO [32], and taxonomic identifiers in NCBI Taxonomy [33]. Shape Expressions test the structural contract of generated RDF. ELK tests the OWL 2 EL portion of the terminology; it is not presented as a complete OWL 2 DL consistency check. A selected forward-rule materialization supports common query paths, and its coverage is reported separately from standard OWL-Horst closure.

The companion ecological analysis reports the inferential results for the quality-controlled community profiles, laboratory-XRF gradient, rainfall, spatial turnover, PMA experiment, association networks and genome-encoded functional potential [34]. The present paper describes the reusable data products, provenance model, identifier mappings and their executable validation.

Environmental digital twins are normally dynamic systems that assimilate observations and support state estimation, simulation, or forecasting [35–37]. The artifact described here does not yet have those capabilities. It is a versioned dataset and semantic backbone on which such a future system could be built; the deposited version becomes immutable only once it is archived under a persistent identifier. Its immediate purposes are data discovery, provenance inspection, and reproducible reuse. It is designed to support cross-domain querying; the executed and archived query evidence at present is the bounded field-XRF retrieval of Section 4.

2 Methods

2.1 Study area and sampling design

We conducted five sampling expeditions across the Rub’ al-Khali along a transect from the western escarpment (19.15°N, 45.17°E) to the eastern border of Saudi Arabia with Oman (20.71°N, 54.73°E). The 60 ordered sampling sites yield a 1,043-km cumulative path and lie 1,015 km apart end to end. Figure 1 illustrates the altitude profile across this transect. The release workflow regenerates this figure and its cumulative-distance profile directly from the 60-site latitude, longitude and altitude table and requires the image bytes to match the staged manuscript figure.

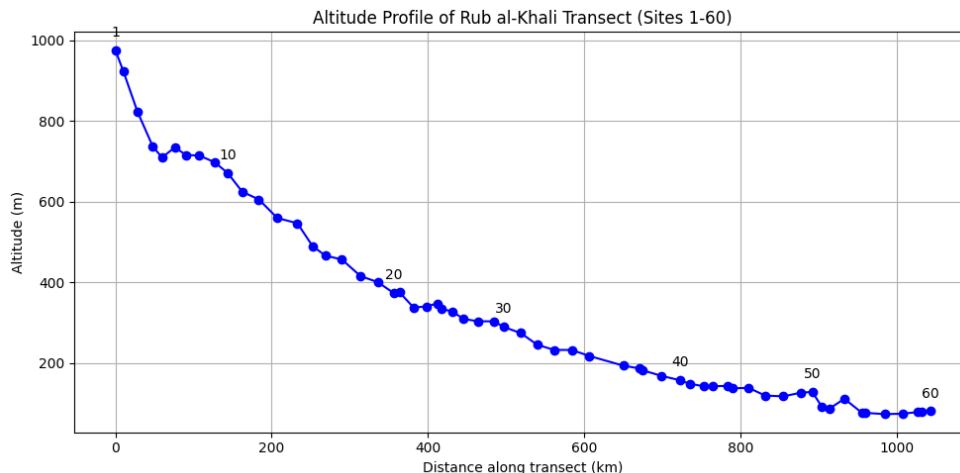


Fig. 1 Altitude profile along the sampling transect across the Rub' al-Khali, from the western escarpment to the eastern border with Oman.

The planned repeated transect encompasses 60 primary sampling sites at predetermined geographic coordinates. The site catalogue also contains 10 named locations of special interest (including water wells, a hot spring and an oil-spill site). The four Trip 1-only numeric identifiers described below are aliases of four named catalogue records and are therefore not added to the catalogue count. The site generator assigns sites 1–60 directly from their numeric labels and assigns the 10 named locations from the versioned bijection in `metadata/samples/site_iri_registry.tsv`. It sorts source sheets and numbers same-label locations from west to east, and fails if the observed named-site set differs from the registry; site identity therefore does not depend on filesystem iteration order.

To capture temporal variability, we conducted expeditions in late winter 2023 (Trip 1), summer 2023 (Trip 2, truncated after site 8 because of extreme temperatures), winter 2024 (Trip 3), summer 2024 (Trip 4), and autumn 2025 (Trip 5). Table 1 describes the records in the field environmental logs rather than implying that every differently labelled record is a distinct physical site. The Trip 3 source sheet contains 60 primary-site records and five named special-interest records from the February 2024 expedition. It also contains 15 appended auxiliary or revisit records whose original workbook dates are 17–20 March 2023; we consequently assign those records to Trip 1 rather than extending Trip 3 into March 2024. Available temperature, atmospheric-pressure and relative-humidity measurements differ by expedition. We acquired X-ray fluorescence (XRF) data through two separate workflows: handheld measurements in the field during Trip 5 and laboratory measurements of archived material from all five expeditions. These workflows differ in setting, specimen coverage and analytical role and are therefore not treated as interchangeable replicates.

The numeric identifiers 61–64 in the Trip 1 source sheets denote genuine Trip-1-only sampling sites, not malformed records. Their regions were inaccessible on subsequent

expeditions. We retain their 36 source rows in release accounting and their available observations in descriptive summaries, but exclude them from inference over the intended repeated-campaign frame of sites 1–60. Their recorded coordinates identify, in order, the existing catalogue individuals *Site water well (location 2)*, *Site road (location 1)*, *Site road (location 2)* and *Site camground*. A versioned alias ledger preserves each numeric source identifier, coordinate, canonical label and canonical IRI. The specimen generator uses that ledger to link the 36 rows to the existing site individuals; it does not mint duplicate site individuals or change their Trip-1-only campaign role. Coordinate matching also resolves the corresponding named Trip 1 field-log record for each of the four sites, so every generated sampling process is linked to the appropriate Trip 1 visit.

Table 1 Coverage of the field environmental logs. Named and auxiliary records are not added to the numeric site catalogue as if they were necessarily distinct physical locations.

Expedition	Season	Field-log records	Measurements available
Trip 1	Late Winter 2023	1–60; 4 named; 15 auxiliary*	Temp, Press, Hum.
Trip 2	Summer 2023	1–8	Temp.
Trip 3	Winter 2024	1–60; 5 named	Temp, Press, Hum.
Trip 4	Summer 2024	1–60	Temp, Hum.
Trip 5	Autumn 2025	1–60; 2 named	Temp, Press, Hum.

*The 15 records are appended to the legacy Trip 3 worksheet but are dated March 2023 in the original workbook and are curated as Trip 1 auxiliary/revisit observations.

The supplementary environmental table is generated from the immutable field sheets through an exact-cell correction ledger. In the Trip 2 sheet, the eight values 34.5–41.9 occur in the source humidity column while the temperature column is empty; their sequence and the field note that the trip ended because of extreme temperature support the pre-existing RDF-generator interpretation of these values as temperature. We therefore move those values to temperature and do not infer Trip 2 pressure or humidity. The Trip 3 site 21 humidity of 31.321% is retained because the original workbook records the fraction 0.31321. The same workbook supplies the March 2023 dates for the 15 appended auxiliary/revisit records; the ledger restores those dates after a later commit changed only the year to 2024 without source evidence. By contrast, the Trip 5 site 40 source humidity of 194% is outside the physical range. Until a primary source establishes a correction, the curated value remains missing and the raw value is preserved only in the correction ledger. The generator fails if a source cell no longer matches its ledger entry or if any retained measurement lies outside its declared physical range. The curated table retains all 274 nonblank records. The graph generator links 259 primary records and the auxiliary record with the exact catalogue label 19.5, yielding 260 site visits with 260 temperature, 187 pressure and 251 humidity values. The other 14 auxiliary/revisit labels denote offset or special-purpose locations and remain table-only; coincident coordinates are not treated as evidence that they are identical to a catalogue site.

At each site, we collected samples from three distinct soil compartments: bulk surface soil (0 cm depth), bulk subsurface soil (5–10 cm depth, hereafter “deep”), and subsurface soil collected near roots (5–10 cm depth, hereafter “root-adjacent”). The companion ecology analysis calls the same 5–10 cm bulk compartment “shallow subsurface”; in this resource and its graph, the legacy label “deep” and the class label “Deep Soil Sample” denote that same source compartment and do not imply a greater sampling depth. We obtained three co-located field replicates per compartment by placing soil into sterile tubes, either by scooping soil directly into the tube or using sterile plastic spoons. We preserved samples at -20°C in the field using an Engel MR040F portable freezer. At the end of each expedition, we transferred samples to an -80°C freezer in the laboratory. These replicates are not independent sites. Companion ecological analyses aggregate or block them by site, campaign and position before site-level inference.

Most canonical bacterial-community profile identifiers encode the expedition, numeric site, compartment and field replicate. Trip 1 has no prefix; Trips 2–5 use T, F, S and V, respectively. Compartment codes are S (surface), D (5–10 cm bulk soil) and PR (root-adjacent soil), followed by r1–r3. Trip 5 suffixes 0, T, R and RE denote extraction, library-preparation or re-extraction events rather than additional field replicates. The release ledger preserves exceptions and links profiles to source records instead of relying on string parsing alone.

The source ledger identifies 34 master-sheet control rows and distinguishes controls, failed extractions, reruns and exclusions by expedition and source row; a further 26 control-labelled rows occur in the sequence submissions. Author-confirmed control facts are frozen in `evidence/controls/control_ground_truth.tsv`. The numbered positive and negative controls are replicate groups, but a reused label is not treated as a globally unique material identifier.

The positive-control design differed by expedition. Trips 1 and 2 used the Zymo-BIOMICS HMW DNA Standard D6322, supplied as purified high-molecular-weight DNA and therefore entering after extraction. Trip 3 used the whole-cell Zymo-BIOMICS Microbial Community Standard D6300 and carried it through extraction. Trip 4 had no positive control. Trip 5 used D6300, but its positive and negative control pair was sequenced by shotgun metagenomics rather than by the 16S assay. D6322 contains seven bacterial genomes at 14 % genomic DNA each and *Saccharomyces cerevisiae* at 2 %; D6300 contains eight bacterial genomes at 12 % each and two yeasts at 2 % each. These manufacturer genomic-DNA fractions are composition specifications, not expected 16S read fractions.

Negative controls included distinct extraction blanks and PCR blanks. EB1–EB18 and Negative1, Negative2, and Negative4–Negative7 are confirmed extraction-blank labels. One EB was included per extraction day, and an extraction day could contain samples from several expeditions; an EB is therefore linked to its extraction batch, never directly to a trip. The extraction-day workbook maps EB1–EB17 to dates and 220 Trip 5 biological profiles; these links do not assign an EB exclusively to an expedition. EB18 and the six Negative-labelled libraries remain unmapped to an extraction day. The Trip 4 workbook records five numbered blanks and one unnumbered blank, but the blank covering a three-day extraction was not sequenced, leaving

all or part of 23 sites without a sequenced blank. Sequence indices distinguish library occurrences but do not establish upstream material identity. The metadata-negative profile `e0323_Ctrl.1_Trip1` is consequently not reclassified from its ambiguous name. No sterile-bag inventory was recovered, so no field-blank instance is invented.

The control registry already assigns PCR-blank roles to three sequenced 16S libraries with explicit PCR or NTC labels (`e0555_PCR.Ctrl.Trip1`, `e0875_NTC.2` and `e8667_PCRCtrl1`) and to one Trip 4 workbook record. The workbook record lists index 275 but has no matched FASTQ dataset. The remaining PCR-control metadata task is laboratory confirmation of any additional reused or generic labels, their PCR batches and the biological libraries amplified in the same batches. Until that evidence is frozen, unresolved label-to-batch claims remain dispositions rather than ground-truth graph edges.

2.2 DNA extraction and sequencing

We utilized 0.9–1.2 g of soil for DNA extraction using the PowerLyzer PowerSoil Kit (QIAGEN, catalog #12855-50) or the DNeasy PowerSoil Pro Kit (QIAGEN, catalog #47014). We homogenized all samples with the PowerLyzer 24 Homogenizer and completed extraction either manually or via the automated QIAcube Connect system. We quantified the eluted DNA using the Qubit HS dsDNA kit.

We amplified bacterial DNA using the Bakt_341F (`CCTACGGGNGGCWGCAG`) and Bakt_785R (`GACTACHVGGGTATCTAATCC`) primer pair [38] to target the V3–V4 region of the 16S rRNA gene. Both primers carried Illumina overhang adapter sequences. We performed amplification using the Multiplex PCR kit (QIAGEN, catalog 206143) with an initial activation at 95 °C for 15 min, 35 cycles of 94 °C for 30 s, 57 °C for 90 s and 72 °C for 90 s, and a final extension at 72 °C for 10 min, and sequenced the resulting pools on an Illumina NovaSeq 6000 platform. We processed all samples in the PI’s laboratory at KAUST.

The reverse primer identity is a correction to the previously recorded protocol, which named the Apprill 806RB primer (`GGACTACNVGGGTWTCTAAT`). A bounded audit of the raw reads resolves the discrepancy against the recorded value: streaming the first 10,000 reads of three Trips 1–4 and three Trip 5 biological read pairs, 56,891 of 60,000 forward reads (94.82%) begin with Bakt_341F and 55,057 of 60,000 reverse reads (91.76%) begin with Bakt_785R, whereas 11 reverse reads (0.018%) match the 806RB sequence. The staged protocol record `metadata/protocols/lib_prep/16S_amplicon_lib_prep.md` carries the corrected pair together with the superseded entry, so the change is auditable rather than silent.

2.3 Sequencing data processing

We processed raw amplicon reads with `nf-core/ampliseq` (v2.14.0) [39, 40], applying the DADA2 algorithm for error correction and denoising and assigning taxonomy against the SILVA 138.2 reference database [41]. The exact command, including the Bakt_341F and Bakt_785R primer sequences, the `--illumina-novaseq` model and the resource configuration, is archived in `metadata/amplicon/`. That directory also contains a portable samplesheet regenerated from the deposited submission records

by `scripts/amplicon/build_portable_samplesheet.py`: 1,268 submitted run rows covering 1,238 distinct sample identifiers, of which 30 carry a second library. Read paths are expressed relative to a single environment variable rather than the absolute cluster paths of the original sheet. No container digest was recorded for the run, so the exact image versions are not pinned.

The canonical Trips 1–5 feature table is the direct output of that pipeline. It is unfiltered: it retains all 351,472 ASVs and still contains the 24 control profiles. No contaminant removal has been applied to it, and none is claimed. A QIIME2 [42] *decontam* branch exists under `analysis/scripts/`, but it operated on a different and smaller table (531 and 975 sample columns), and in the form that survives, the command that would have produced the contaminant table is commented out while a later step consumes that missing artifact. No removal fraction from that branch is therefore defensible, and the previously reported prevalence threshold is not restated as an applied filter. That branch retained ASVs with a minimum total frequency of five reads in at least two samples.

An assay-aware audit following low-biomass control recommendations [43] is instead generated by `scripts/controls/run_assay_aware_control_audit.py`. The primary test pools the 17 extraction-day blanks with frozen links to Trip 5 profiles and uses an uncorrected one-sided Fisher $p < 0.1$, requiring presence in at least two blanks. At the same threshold, requiring only one blank would call 12,051 rather than 351 ASVs. Because one blank represents each extraction batch, within-batch prevalence estimation is impossible. The resulting candidate set is applied only to the 217 canonical biological profiles linked to those batches; EB18 and the six **Negative**-labelled libraries are characterization-only. Frequency modelling is not run because control DNA concentrations are unavailable. Seven unambiguously positive-labelled 16S profiles from Trips 1–3 in the frozen pre-decontamination table are used to evaluate expected bacterial-genus recovery and exact-ASV occurrence, never to train the contaminant classifier. Their library-to-product assignments are provisional hypotheses from positive labels and the confirmed trip-level design. Expected bacterial genera accounted for 5.85, 12.33, 31.05, 0, 0.05, 100 and 99.99% of reads in `e0553_Ctrl1_2`, `e0554_Ctrl1_3`, `e0872.TMC1`, `e0874.Positive`, `e8294.FPosCtrl11`, `e8661.FPosCtrl12` and `e8665.FPosCtrl13`, respectively. Only the final two met the declared descriptive mock-dominated threshold of at least 95%. Manufacturer genomic-DNA percentages are not expected 16S read percentages. The primary table remains unfiltered; `scripts/controls/run_control_ecology_sensitivity.sh` rebuilds the ecological inputs and 25 tracked headline metrics across 11 claims solely as a bounded sensitivity analysis (Section 5).

Genome-encoded functional potential used for the companion analysis was predicted with PICRUSt2 v2.4.1; its execution manifest, with the exact commands, the PIC hidden-state method with edge exponent 0.5 and seed 100, the NSTI cut-off of 2.0, the minimum-read threshold of 100 and the recorded resources, is archived at `metadata/functional/PICRUSt2.md`. The predictions came from two separately executed input sets: the Trips 1–4 ampliseq output and the corrected Trip 5 ampliseq output, whose input universe contained 330,830 ASVs. They were not generated from the later 351,472-ASV combined canonical table.

All deterministic metadata corrections are recorded without overwriting the source values. In records F46Dr2, S46Dr2 and V46Dr2, the value `NEBNext` had shifted into the site column while the other populated cells in those rows remained aligned. The confirmed correction ledger restores site 46, records `NEBNext` as the displaced WGS library-preparation kit value and retains the original cell value and rationale. The repair does not infer that WGS was performed: sequencing status and run availability remain separate fields and are unchanged.

2.4 Companion-analysis derived inputs

The staged release candidate contains the exact derived inputs needed to re-execute the reported downstream companion analyses. The `metadata/relic-dna/PMA_ASV_table.tsv` matrix has 18 analytical columns representing nine Trip 5 treated–untreated aliquot pairs, with controls retained separately. The `metadata/metagenome/coverm_profiles.tar.gz` archive contains 150 sample-level genome-abundance profiles, and `metadata/metagenome/eq.emapper.annotations.gz` supplies the eggNOG annotations used to construct the encoded-function matrix for the 990 genomes shared with those profiles. The `metadata/metagenome/measured_function_inputs.tar.gz` archive contains the six exact source tables used for the genome-derived versus PICRUSt2 comparison and metabolic-marker summary.

These are derived statistical inputs, not a replacement for the underlying sequence data. The PMA table and measured-function archive directly support their named endpoints; the encoded-function null also requires the bulk-listed eggNOG file. These inputs do not reproduce upstream shotgun or PMA read processing, metagenome assembly, binning, dereplication, annotation or CoverM generation. Raw shotgun and PMA reads and the full metagenome-assembled-genome catalogue therefore remain separate deposition requirements.

2.5 Taxonomy normalization and mapping

Amplicon classifications were assigned against SILVA 138.2. We retain the original classifier lineage and ASV identifier as the primary evidence. The combined Trips 1–5 export contains two source encodings: the seven SILVA rank fields alone, or those seven fields followed by a separate species-assignment field and a numeric confidence. QIIME artifact provenance identifies the latter as an appended species assignment rather than an eighth rank. We therefore normalize only the ordered domain–species fields. Rank prefixes are removed, empty and unclassified values are represented explicitly, and the separate species assignment is retained in a provenance ledger but is never promoted into the asserted lineage. This rule also preserves, without silently adjudicating, supplementary species calls that conflict with the classifier’s species field.

A fail-closed normalization step evaluates the historical SILVA-to-NCBI candidates against a pinned NCBI Taxonomy release: the OBO Foundry NCBITaxon release 2025-03-13 (<http://purl.obolibrary.org/obo/ncbitaxon/2025-03-13/ncbitaxon.owl>), which is built from NCBI organismal taxonomy version 2025-03-01. The exact artifact consumed is 1,646,833,334 bytes with SHA-256 5238897945e50ceb9007a1f96f40d1e8774456c3a0d691292fe7e8e9b5b1287d; the checksum identifies the file and the release identifier states which version it is.

The source ontology IRI, version IRI, `owl:versionInfo`, NCBI source date, source checksum and derived-subset checksum are bound machine-readably in `ontology/ncbitaxon_release_identity.json`; the historical retrieval date was not recorded and remains an explicit null. An NCBI identifier is retained only when it resolves, has a compatible rank and label or documented synonym, and its ancestor chain is compatible with all retained higher-rank identifiers in that source lineage. A missing, ambiguous or incompatible candidate is replaced by a deterministic project class scoped to the complete truncated source lineage and rank. It is not replaced by the nearest ancestor’s identifier: doing so would conflate a named child with its broader ancestor. Each decision records the source lineage and rank, candidate identifier, validation outcome, final identifier and reason. The output mapping must cover every canonical lineage represented in both the Trips 1–5 taxonomy export and feature table; otherwise generation stops.

Existing project identifiers are retained only when all explicit historical uses share one normalized rank, label and nearest explicit parent context. The audit is repeated against the corrected parent IRIs to a fixed point; a reused identifier that acquires more than one corrected context is replaced in every affected use by contextual classes. Finally, every occurrence of the same rank-truncated source lineage must have one final IRI, label and mapping status regardless of which lower-rank descendants occur beneath it. Conflicting historical evidence is therefore contextualized rather than resolved according to file order.

The mapping builder emits corrected JSON and tabular mappings, a decision ledger, a clean project taxonomy module, an audit report and a manifest containing input and output byte counts and SHA-256 checksums. The abundance-ABox generator accepts only a passing, checksum-matched manifest. It aggregates independently by profile, rank, final identifier and truncated source lineage, thereby preserving distinct classifier contexts even when they share an accepted NCBI identifier. No global label or parent is inferred by overwriting one source lineage with another.

Cross-resource alignment to GTDB [44] and iNaturalist was evaluated separately from this SILVA-to-NCBI layer, in an earlier exploration whose candidate and discordance tables are not part of the release and were not regenerated for it; the counts they reported are therefore not restated here. The conclusion that the exploration supports is a negative one and does not depend on those counts. Homonyms, *Candidatus* names, placeholder taxa, GTDB split taxa, rank conflicts and database-version drift can all produce identical labels for non-identical concepts. Label agreement and the former parent-voting heuristic therefore do not establish extensional identity.

The canonical release modules contain no `owl:equivalentClass` assertions generated from those lexical candidates. Unreviewed GTDB and iNaturalist candidates are excluded from the asserted crosswalk, and no mapping precision or recall is claimed because a stratified manual gold standard has not been completed. This conservative representation permits queries over the original SILVA lineage and validated NCBI classes without turning lexical similarity, a shared label or a supplementary species call into unsupported taxonomic identity.

2.6 XRF acquisition, processing and reconciliation

Field-XRF.

During Trip 5, researchers used a Vanta handheld instrument to perform measurements in the field. The source log contains 106 entries across 59 sites; 71 complete sessions at 58 sites were retained, and nine of those sites have repeated complete sessions. The field table retains the instrument test identifier, site identifier, reported analyte values and instrument uncertainty columns. Project records identify these as deep-soil measurements, but the instrument log itself does not encode a physical-sample, compartment or depth identifier. Field/laboratory linkage is consequently site-level, not a same-aliquot comparison. The exports encode the `Geochem(3-Beam)` method and `NORMAL` mode.

Laboratory-XRF.

Archived material was subsequently measured in the laboratory for all five expeditions. The canonical inputs contain 547 selected sample records from Trips 1–4 and 178 selected records from Trip 5 (725 records in total), and all 725 are used in the analyses. A former 158-row Trip 5 table and the resulting 705-row cross-trip table are retired provenance artifacts; neither is a current analytical input. Trip 5 workbooks encode the `Fast Screening-He8mm` method, `He` mode, 8-mm diameter and oxide material. Equivalent acquisition metadata have not been recovered for Trips 1–4.

Aggregation audit.

The provenance audit reproduced all 13,390 reported Trips 1–4 cells under the current largest-positive-value rule and all 4,293 reported Trip 5 cells under the parser’s different last-reported-value rule. Fifteen rare analytes present in the Trips 1–4 source rows are absent from the current wide-table schema. Neither rule is yet treated as instrument-prescribed: a larger value is not necessarily more accurate, and zero means only “not reported positive” while detection-limit semantics remain unavailable. The audit compares maximum-positive, mean-positive, median-positive, first-reported, last-reported and workbook-section-specific candidates, but selecting these source-specific policies does not establish instrument accuracy. The release preserves and discloses the Trips 1–4 maximum-positive and Trip 5 last-reported policies instead of implying that they are a common calibrated procedure. Alternative summaries are retained as a sensitivity audit; they are not substituted into the primary table.

Chemical-identifier audit.

All instrument channels are declared once in `config/codes/xrf_chemical_mapping.yml`. Element channels were mapped to neutral-atom records; oxide channels were mapped only when the pinned source contained a neutral formula-matching record. The executable audit streams the configured identifiers from pinned ChEBI v247, rejects missing or deprecated records and mismatches in formula, charge or atom-versus-oxide type, and performs the same checks against a dated PubChem property snapshot. ChEBI formulae that use dot notation are compositionally parsed. Two documented cases are handled explicitly: ChEBI does not assert formula or charge

metadata for its exact-label magnesium-oxide record, and P_4O_{10} is accepted for the P_2O_5 reporting channel only as an empirical oxide-equivalent. No presence of a particular oxide molecule or mineral is inferred. Unsupported mappings remain null; LE remains an unmapped instrument pseudo-analyte.

For a descriptive cross-workflow diagnostic, repeated field sessions were reduced by their site median and matched to the Trip 5 laboratory deep-soil record at 58 sites. This is not same-aliquot validation. Across the reported analytes, rank agreement was generally modest and field/laboratory ratios were analyte-dependent. The canonical analytical table and regenerated knowledge graph use all 725 laboratory records while keeping the field sessions as a separate observation type. No cross-workflow calibration is inferred, and no ecological interpretation assumes that field and laboratory measurements have the same scale or are interchangeable.

2.7 Archived-soil pH acquisition and admission

We measured archived dry soil after sieving below 2 mm and mixing without fine grinding. Each assay used an independent 1 g scoop suspended in 2.5 mL 0.01 M $CaCl_2$. Measurements were made with an accumet AB315 and validated Probe 3 under automatic temperature compensation. Each measurement session recorded a two-point pH 7/10 calibration, electrode slope and pH-10 buffer read-back; admission required calibration marked yes, slope 95–102 % and a read-back within 0.1 pH units of nominal. Historical probe-comparison and reference-soil sheets document method validation but are not biological observations.

All measurements available when laboratory follow-up ended on 3 August 2026 were frozen as `EQ-PH-SHARED-v1.0.0`. This is the sole pH version used by this descriptor and the companion ecology paper. The source workbook and version manifest are under `metadata/samples/ph/versions/EQ-PH-SHARED-v1.0.0/`; the normalized audit tables are under `metadata/ph/`. We reconciled each trip-sheet row to the sample ledger and existing specimen IRI. A row entered the graph only when it also carried a numeric pH value between 0 and 14 and a source date represented as an Excel date, ISO date or day-first DD/MM/YYYY date that did not fall after the cutoff. Unparseable dates and dates that resolved beyond the cutoff were quarantined. We neither inferred missing quality-control fields nor imputed unavailable values.

The audit accounts for all 1,168 source rows. It admits 712 measurements, retains 356 rows without a measurement, marks 45 as depleted, quarantines 36 with ambiguous dates, and excludes 19 numeric rows whose recorded quality-control fields are incomplete. The measurement campaign is therefore complete as available, not complete for every listed specimen. For Trip 4, `S28Sr1` was missed during preparation and `S57Dr1` was depleted. Future corrections or recovered measurements require a new version; they cannot replace this frozen source.

The workbook has no session identifier. We reconstructed 29 session individuals from unique combinations of trip, accepted measurement date, electrode slope and pH-10 read-back. These groupings preserve the available quality-control context but are not directly recorded laboratory batch IDs.

2.8 Environmental and climate data

We retrieved historical climate data from the Open-Meteo archive endpoint [29] for each recorded field location in two separately configured acquisitions. The monthly acquisition covered 1 January 2022 to 20 January 2026 and requested daily mean temperature, total precipitation, rainfall and mean relative humidity before aggregation to monthly means and sums. A separate daily table covered 1 January 2022 to 1 February 2026 and requested temperature, total precipitation and rainfall, but not humidity. The regenerated graph contains 12,936 monthly climate measurement records encompassing temperature, total precipitation, rainfall and relative humidity. The annual source fields contain 636 values for temperature, total precipitation and rainfall; exact site-key resolution retains 582 values in the graph and quarantines 54 values attached to unresolved field-notebook labels.

Retrieval keyed on the `Site` column of the per-trip geodata sheets, which is a field notebook column rather than an integer primary key. The site catalogue contains 70 sampling-site individuals: 60 numbered sites and ten named locations. Applying the same label resolution as the RDF generator, we indexed each individual by its `rdfs:label` and, for labels beginning *Site*, by the remainder. Under this rule, 66 of the 81 distinct fetch keys resolve to a site individual: all 60 numbered sites and six named ones addressed by their bare label (*oil spill*, *hot spring*, *Camell1*, *plant gradient*, *19.5* and *camground*). That is the denominator behind the released counts: 66×49 months $\times$ four variables gives 12,936 monthly records.

The remaining 15 keys resolve to no site individual. Thirteen carry a numeric stem with an unrecorded offset or suffix (*27+*, *47+grass*, *52AD* and similar) and two are free-text locations (*road*, *water well*) that do not match a catalogue label. A near-site label is deliberately not mapped onto its stem: *47+grass* records a reading taken near site 47, not a reading of site 47, and the offset was never written down. `scripts/climate/curate_climate_keys.py` resolves every key against `ontology/rubalkhali_sites.owl`, records the disposition and resolved site IRI in `climate_fetch_key_ledger.tsv`, and writes site-resolved projections of the monthly (3,234 rows retained, 735 quarantined) and daily (98,538 retained, 22,395 quarantined) tables. The same exact-key rule retains 194 of 212 geodata rows with annual fields (582 of 636 values) and quarantines the remaining 18 rows (54 values); annual records are not coordinate-matched onto a catalogue site.

Acquisition provenance is only partly recoverable. The endpoint, request parameters, date window and input and output checksums are recorded in `climate_acquisition_frozen.json`. The raw response bodies, the per-request timestamps and any upstream model or dataset version were never captured for the archived tables, so the historic acquisition cannot be reproduced byte-for-byte and is not claimed to be. The acquisition code has since been given an explicit frozen date window and writes response snapshots, which makes a **future** retrieval re-derivable from its own snapshots; it does not retrospectively reconstruct the archived one, and the first such retrieval still has to contact the service. The citations to MIXS/MIMARKS and STREAMS provide reporting context; this package does not claim audited compliance with either standard. A field-level crosswalk and completed reporting checklist

remain pre-submission author tasks. Furthermore, field researchers manually classified each site according to its biome (e.g., desert biome) and specific environmental features (e.g., sand dune, saline pan, gravel, desert oasis). We subsequently mapped these classifications to the Environment Ontology (ENVO) [32].

2.9 Schema validation

To ensure structural integrity within the RDF graph, we used Shape Expressions (ShEx) [45]. ShEx provides a formal structural schema language that validates the presence, cardinality, and datatype of RDF properties. We defined specific ShEx shapes for all core entity types in the knowledge graph. These entity types include sampling sites, soil samples, DNA extracts, sequencing datasets, measurements, quality control records, laboratory control materials, control roles, laboratory batches, control sequence occurrences, archived-soil pH processes, values and sessions, and taxonomy mappings.

The validation workflow binds node selectors (for example, all instances of the sampling-site class) to their ShEx definitions using a Shape Map and fails if it detects a missing mandatory property (for example, site coordinates) or an invalid datatype. The site, specimen, environmental-measurement, DNA, XRF, sequence-record and QC modules passed their assigned shapes. The separate control module also passed its shape and scientific-invariant checks, including unique role bearers, paired input/output and role-realization directions, no direct control-to-trip or batch-to-trip link, and no positive control in the contaminant-training set. The multi-gigabyte taxonomy ABox was excluded from this memory-bounded ShEx run; the exact generated file was instead parsed in full with an independent Turtle parser and checked by a streaming validator over every classification process–dataset–quality–value relationship and identifier range. Passing establishes conformance to the declared structural contract appropriate to each stated validator; it does not establish source accuracy or biological validity, and the streaming check is not reported as an unexecuted ShEx result. The pH module was validated separately over all 712 process–quality–value sets and 29 sessions. Its positive Shape Map passed, while a single-defect fixture lacking UO:0000196 was rejected as expected. The Turtle and RDF/XML serializations were also required to be graph-equivalent. The validator uses the Apache Jena ShEx library (version 4.10.0) invoked from custom Groovy scripts.

2.10 Reasoning coverage

We use ELK [46] for a bounded consistency test over the loaded modules and the OWL 2 EL constructs it supports. ELK is not complete for arbitrary OWL 2 DL constructs, including universal restrictions and inverse properties used by imported SIO modules. Consequently, a successful ELK run establishes neither full OWL 2 DL consistency nor the absence of every possible ABox contradiction. In the current memory-bounded LITE run, ELK reported the merged included modules as consistent. The run did not precompute the class hierarchy and therefore did not test whether

uninstantiated named classes were unsatisfiable. Large reference and taxonomy modules were excluded, so this result is bounded consistency of the included modules rather than whole-graph logical consistency.

2.11 Reproducible workflow

Release generation, validation and manuscript-derived statistics are orchestrated from one Nextflow entry point (`main.nf`). Separate local/workstation and IBEX Slurm profiles change execution infrastructure without changing process definitions. The executed workstation run generated the source-provenance snapshot, sample-flow ledger, release reconciliation, XRF audit, ecology analyses, resolution-matched functional null and network analysis. In an isolated task it also regenerated the site, environmental-measurement, sample, XRF, DNA, sequence-record and QC modules from their declared sources and required every byte stream to match the staged release candidate. The same generated-module stage rebuilds the shared pH module from its frozen workbook, admission policy and specimen graph. The separate `scripts/analysis/run_ph_shared_v1.sh` driver reproduces the pH ecology models, version comparison, manuscript values and full ShEx validation. The corrected taxonomy mapping and full taxonomy ABox were then generated using that fresh sequence-record module, and the knowledge-graph gates consumed those exact upstream artifacts rather than the checkout’s prebuilt ABoxes. The curated base TBox/RBox is a versioned workflow input, not a data-derived output. The paper build is a separate leaf stage so that rendering can run on a TeX-enabled host without changing any scientific result.

2.12 Knowledge representation

We developed a formal knowledge base to integrate the different types of data generated in this project. The semantic layer encodes the recorded provenance chain from physical sampling to digital data products, supporting cross-domain queries and interoperability through shared vocabularies.

2.12.1 Architecture

The knowledge base is constructed in Description Logic [47] using a modular architecture that strictly separates the terminology (Terminological Box, TBox, or ontology) from the observed facts (Assertion Box, ABox, or domain knowledge). The TBox uses OWL 2 [48], while the ABox contains the instance data generated from the expeditions. The deployment uses separate named graphs for the terminology, asserted expedition data, taxonomy references and derived assertions. The current module-level entity counts are reported in Section 3; no historical whole-graph count is used as a proxy for the regenerated package.

2.12.2 Prefixes used in listings

For brevity, the Turtle and SPARQL listings throughout the paper use the prefix declarations of Table 2; these declarations should be prepended to any listing that the reader wishes to execute against a local or deployed graph. For example, the

full IRI `https://rubalkhali.science/kb/RAK.P100001` is written `rak:P100001`, and `http://semanticscience.org/resource/SIO_000291` is written `sio:000291`. The separate `rak_kb:` prefix covers the same knowledge-base namespace without the `RAK_` local-name prefix. It is needed for one legacy individual, the PowerSoil extraction protocol `rak_kb:DNA_ext_powersoil`, which is the only individual in the released graph minted without the `RAK_` prefix; every other protocol individual (for example `rak:L000010`, `rak:L.Surface.Lab`) follows the prefixed convention. The inconsistency is recorded here rather than silently corrected because changing the IRI would break the checksums of the already-validated modules.

Table 2 Prefix declarations used in all Turtle and SPARQL listings.

Prefix	Namespace
<code>rak:</code>	<code>https://rubalkhali.science/kb/RAK_</code>
<code>rak_kb:</code>	<code>https://rubalkhali.science/kb/</code>
<code>sio:</code>	<code>http://semanticscience.org/resource/SIO_</code>
<code>owl:</code>	<code>http://www.w3.org/2002/07/owl#</code>
<code>pato:</code>	<code>http://purl.obolibrary.org/obo/PATO_</code>
<code>uo:</code>	<code>http://purl.obolibrary.org/obo/UO_</code>
<code>envo:</code>	<code>http://purl.obolibrary.org/obo/ENVO_</code>
<code>ncbitaxon:</code>	<code>http://purl.obolibrary.org/obo/NCBITaxon_</code>
<code>obo:</code>	<code>http://purl.obolibrary.org/obo/</code>
<code>rdf:</code>	<code>http://www.w3.org/1999/02/22-rdf-syntax-ns#</code>
<code>rdfs:</code>	<code>http://www.w3.org/2000/01/rdf-schema#</code>
<code>xsd:</code>	<code>http://www.w3.org/2001/XMLSchema#</code>
<code>geo:</code>	<code>http://www.opengis.net/ont/geosparql#</code>
<code>dcterms:</code>	<code>http://purl.org/dc/terms/</code>

2.12.3 Ontology engineering and design patterns

We use the Semanticscience Integrated Ontology (SIO) [30] as the upper-level framework. SIO provides general-purpose classes and relations applicable across scientific domains, supporting semantic interoperability with other SIO-based datasets. We defined all domain-specific classes as subclasses of SIO classes, and all introduced relations are sub-relations of SIO relations. We mapped classes to SIO and maintained full semantic mappings in our public repository (see Section 7).

The canonical terminology module (`ontology/rubalkhali.owl`) declares 333 classes: 297 project-local `RAK_` classes and 36 referenced classes (32 SIO and four PATO). It also declares 20 object properties (11 project-local and nine SIO properties) and 35 datatype properties (34 project-local and one SIO property). These figures are declaration counts from the staged canonical module; they are not counts of the imported ontologies or of whole-graph OWL axioms. The release workflow recounts the distinct serialized declaration IRIs, fails if they differ from these values and records the result in `evidence/semantic-validation/ontology_declaration_audit.json`.

Measurement pattern

We model a quantitative measurement using SIO's ontology design pattern for measurement processes: a measurement is a process that generates a value for a specific quality of a target entity.

$$\text{MeasurementProcess} \sqsubseteq \text{Process} \quad (1)$$

$$\text{Quality} \sqsubseteq \text{Attribute} \quad (2)$$

$$\text{Quantity} \sqsubseteq \text{InformationContentEntity} \quad (3)$$

$$\text{MeasurementProcess} \sqsubseteq \exists \text{hasTarget}.\text{Entity} \sqcap \exists \text{hasOutput}.\text{Quantity} \quad (4)$$

$$\text{Quantity} \sqsubseteq \exists \text{hasValue}.\text{xsd:double} \sqcap \exists \text{hasUnit}.\text{Unit} \quad (5)$$

$$\text{Quantity} \sqsubseteq \exists \text{isMeasurementValueOf}.\text{Quality} \quad (6)$$

We identified all units with classes from the Units Ontology (UO) [49]. Qualities are expressed using the Phenotype and Trait Ontology (PATO) [50] and linked to SIO as subclasses of the *Quality* class. To illustrate this pattern, we provide an example represented in Turtle format. The example (Listing 1) demonstrates an environmental temperature measurement taken at a sampling site.

Listing 1 Turtle representation of an environmental temperature measurement.

```
rak:P000001 a rak:0000006 ; # Measurement Process
  sio:000068 rak:3000001 ; # isPartOf (site visit)
  sio:000139 rak:A000001 ; # hasAgent
  sio:000132 rak:D000002 ; # hasParticipant (device)
  sio:000229 rak:4000001 . # hasOutput (Value)

rak:3000001 a rak:0000003 ; # Site Visit
  sio:000291 rak:1000001 . # hasTarget (Sampling Site)

rak:4000001 a rak:0000010 ; # Quantity
  sio:000221 uo:0000027 ; # hasUnit (degree Celsius)
  rak:2000003 "20.7"^^xsd:double ; # hasValue
  sio:000215 rak:5000001 . # isMeasurementValueOf (Quality)

rak:5000001 a pato:0000146 ; # Temperature Quality
  sio:000011 rak:1000001 ; # isAttributeOf
  sio:000216 rak:4000001 . # hasMeasurementValue
```

In words, an instance `rak:P000001` of a measurement process is part of the site visit `rak:3000001`, which in turn has the sampling site `rak:1000001` as its target, and the process produces as output the quantity `rak:4000001`. The target edge on the process itself is not asserted: the terminology declares the property chain `isPartOf o hasTarget  $\sqsubseteq$  hasTarget`, so `rak:P000001 sio:000291 rak:1000001` is entailed rather than stored. A query that must run without reasoning should therefore traverse the visit explicitly. The visit individual is not redundant bookkeeping: it carries the campaign and the exact date-time at which the reading was taken, so repeated visits to one site across campaigns remain distinguishable. That quantity carries the literal value 20.7, is reported in degrees Celsius (`uo:0000027`), and is the measurement value of a temperature quality `rak:5000001`, which is in turn an attribute of the

same site. In SIO, `sio:000215` is the value-to-quality relation and `sio:000216` is its quality-to-value inverse. Reading the chain in either direction therefore connects the numeric reading to both the measured site and the abstract quality that it quantifies.

Archived-soil pH uses the same four-individual measurement pattern without treating an archived-soil assay as a field-site observation. The existing specimen bears an acidity quality (PATO:0001842) through `sio:000008`. A value individual is typed as both SIO measurement value (SIO:000070) and pH (SIO:001089), carries its numeric literal through `sio:000300`, and has unit UO:0000196. The value points to the quality with `sio:000215` and to the measuring process with `sio:000232`; the quality points back with `sio:000216`. The SIO measuring process (SIO:001054) consumes and targets the specimen, outputs the value, is specified by the CaCl_2 protocol, has the instrument as a participant, and is part of a dated measurement session. Source-workbook and session provenance remain connected to the process. Rows that fail source, date, specimen or quality-control admission remain in the audit table and are not asserted as measurements.

Sampling site and biome assignment pattern

Each physical sampling location is represented as a persistent individual that is grounded in the Environment Ontology (ENVO) by linking to a biome and to one or more environmental features. An existential restriction requires a biome filler, while a universal restriction constrains every such filler to the ENVO biome class. Under OWL's open-world semantics this does not make the asserted ENVO biome class the unique possible filler. Because ShEx shapes match triple patterns and the biome and feature assignments are encoded as class restrictions rather than as simple triples, no shipped shape tests biome completeness. Closed-world completeness is instead enforced by a separate query gate, `scripts/validation/validate_site_biome_completeness.py`, which fails if any `rak:0000002` individual lacks a `rak:2000001` or `rak:2000002` restriction.

SamplingSite $\sqsubseteq$ Site (7)

SamplingSite $\sqsubseteq \exists \text{hasBiome}.\text{ENVO:Biome} \sqcap \forall \text{hasBiome}.\text{ENVO:Biome}$ (8)

SamplingSite $\sqsubseteq \exists \text{hasEnvironmentalFeature}.$
ENVO:EnvironmentalFeature (9)

Listing 2 Turtle representation of a sampling site grounded in ENVO.

```
rak:1000001 a rak:0000002 ; # Sampling Site
  rdfs:label "Site 1" ;
  geo:asWKT
    "POINT(45.166511388888885 19.149770555555556)"^^geo:wktLiteral ;
  a [ a owl:Restriction ;
      owl:onProperty rak:2000001 ; # hasBiome
      owl:someValuesFrom env:01000179 ], # desert biome
    [ a owl:Restriction ;
      owl:onProperty rak:2000002 ; # hasEnvironmentalFeature
      owl:someValuesFrom env:01000018 ]. # gravel
```

In words, the sampling site `rak:1000001` carries a label and a WKT geometry as ordinary data properties, and its biome and environmental feature are encoded as OWL class restrictions on the individual rather than as direct object-property assertions. This matters for reuse: a query of the form `?s rak:2000001 ?biome` returns no rows against the asserted graph. A reader must either match the restriction structure, as in

Listing 3 SPARQL pattern that recovers the biome of a sampling site from the asserted restriction structure.

```
?site a rak:0000002 ;
      a [ owl:onProperty rak:2000001 ;
          owl:someValuesFrom ?biome ] .
```

or query a materialized graph in which the restriction has been expanded. The combined existential and universal restriction on `hasBiome` (Equation 7) does not identify a unique biome filler. Under OWL’s open-world semantics it requires at least one `hasBiome` value in the named ENVO class and constrains every such value to that class; it does not entail that the site has only one biome filler. The recorded environmental feature for Site 1 is the ENVO material *gravel* (`envo:01000018`); features are drawn from the project’s own code table, which maps free-text field descriptors to ENVO identifiers, and a feature is therefore a recorded surface descriptor rather than a curated landform classification.

Laboratory-control pattern

A laboratory control is represented as an occurrence-specific material entity, not as a property of an expedition and not as the string used to label a library. The material bears its own instance of a positive, negative, extraction-blank or PCR-blank role through `sio:000228` (*has role*); the inverse `sio:000227` is asserted explicitly. A laboratory process takes the material as input and target through `sio:000230` and `sio:000291`, produces the derived material or dataset through `sio:000229`, and realizes the role through `sio:000355`. The inverse input, target, output and realization directions are also asserted because the deployed validation and query paths do not assume complete inverse-property materialization.

An extraction or PCR process is linked by `sio:000068` (*is part of*) to its processing batch. Biological materials processed in the same batch retain their own expedition provenance; neither the blank nor the batch receives a direct trip link. This is essential here because an extraction day could contain samples from several trips. Whole-cell D6300 and purified-DNA D6322 controls enter the workflow at different processes, so the graph can distinguish extraction and lysis evaluation from PCR, library and sequencing evaluation. A product is linked to its manufacturer composition specification by `sio:000339` (*is specified by*), with `sio:000338` asserted in the reverse direction. Each expected taxon is both a promoted specification-to-taxon fact and a reified, evidence-bearing expected-taxon assertion. Manufacturer genomic-DNA percentages, their unit IRIs and composition basis remain in the normalized table rather than being represented as expected 16S read fractions in RDF.

Uncertain identity, batch membership and assay applicability are represented as SIO statements linked to source evidence with `sio:000772`; they carry an explicit

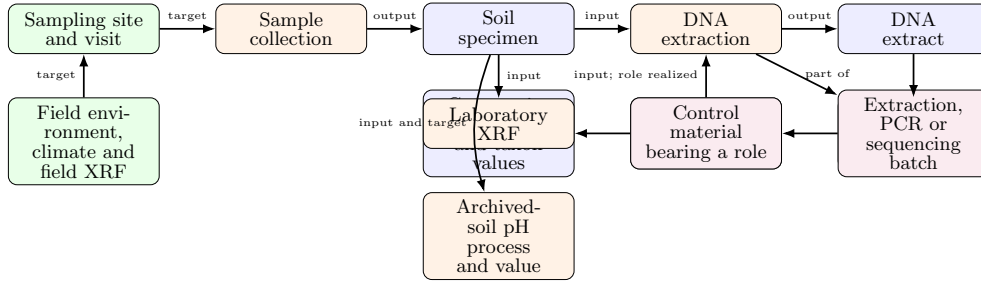


Fig. 2 Provenance and module architecture. SIO input, output, target, role-realization and part-of relations connect field context to specimens, molecular products and digital observations. Field environment, climate and field XRF target a site; laboratory XRF consumes a specimen. Control materials bear occurrence-specific roles realized in laboratory processes and connect to processing batches rather than directly to expeditions. Archived-soil pH consumes and targets the specimen and produces a protocol-bound value. Separate ABox modules serialize these domains while retaining the same site, specimen and process identifiers.

confirmed, provisional, unresolved or inapplicable status. Each material occurrence has a separate role individual, and reused labels are represented as aliases, preventing functional role relations from merging physical controls. No sterile-bag role class or material instance is asserted. An unresolved inventory disposition records that bag identifiers and sequence occurrences were unavailable; the model can be extended if an inventory is recovered. A machine-checked D6300 composition and expected-taxon example is provided in Supplementary Section S1.

Patterns specified in the supplement

The measurement, sampling-site and laboratory-control patterns above are reproduced here because they carry contracts a reuser must know: the value–quality direction, the restriction encoding of biome and environmental feature, and the separation of control material, role, process and evidence. The remaining six patterns (sampling, protocol execution, taxonomic abundance, XRF geochemistry, the sequencing chain and sequencing quality control) are specified with their axioms and machine-checked worked examples in Supplementary Section S1. Full generator sources, the complete class and property mappings and the shape files are distributed with the package (Section 3.8).

2.12.4 OWL and RDF implementation

Figure 2 summarizes how the modules connect field context, specimens, molecular products, digital observations and laboratory controls. We generated the OWL/RDF implementation using custom Groovy scripts utilizing the OWL API 5 [51]. The knowledge base employs a modular design, separating the base ontology, sites, samples, measurements, XRF data, DNA extraction records, sequencing metadata, quality control metrics, laboratory controls, archived-soil pH, and taxonomic abundances into individual ABox modules [52].

We use OpenLink Virtuoso [53] for RDF storage and query answering. In a development benchmark, a project-specific forward-rule program applied selected

RDFS/OWL rules for subclass, subproperty, domain, range, inverse, symmetric, equivalence and transitive-property assertions. The program used three scheduled passes, including only a one-hop transitive pass. It did not demonstrate convergence to a fixed point and did not implement arbitrary OWL property chains. We therefore describe this output as a *selected rule expansion*, not as complete OWL-Horst [54] closure.

The current staged package contains the regenerated asserted modules but no validated materialized whole-graph artifact. We consequently report exact module-level entity counts rather than historical asserted or inferred triple counts. The selected rule expansion has not been demonstrated to reach a zero-delta fixed point and does not implement arbitrary property chains; neither complete OWL-Horst coverage nor a complete inferred-graph count is claimed.

2.12.5 Schema definition

We used Shape Expressions (ShEx) [45] to define structural schemas for the core entity types. While OWL provides open-world reasoning and inference capabilities, it cannot readily enforce data completeness constraints. We developed ShEx schemas to explicitly validate the graph’s structure, supporting data-quality and conformance checks before loading it into the database.

Our ShEx implementation defines shapes for Sampling Sites, Soil Samples, Environmental Measurements, DNA Extracts, and Sequencing Datasets. These shapes require the declared attributes on conforming nodes. For instance, Listing 4 shows how the schema requires every conforming Sampling Site to be explicitly typed, labelled and georeferenced, while optionally capturing a textual description.

Listing 4 ShEx shape for Sampling Site validation.

```
<#SamplingSiteShape> {
  rdf:type [rak_kb:RAK_0000002] ; # Must be a Sampling Site
  rdf:type . * ; # May have other inferred types
  rdfs:label . ; # Must have a label
  dct:terms:description xsd:string ? ; # Optional description
  geo:asWKT . ; # Must have coordinates
}
```

In words, the shape `<#SamplingSiteShape>` requires that every conforming node carry an explicit type assertion to `rak:0000002` (Sampling Site), an `rdfs:label` and a WKT geometry; it permits any number of additional inferred types and optionally allows a free-text description. Coordinates are mandatory in the shape, in the Methods contract of Section 2 and in the staged copy of the shape; all 70 sampling-site individuals in the released module satisfy it. A site record that omits the mandatory type, label or geometry is rejected before it reaches the triple store, and a deliberately coordinate-free fixture is included in the negative fixture suite (Section 4) to prove that the rejection actually fires. Note that the shape file declares `rak_kb:` for the bare knowledge-base namespace, so the sampling-site class is written `rak_kb:RAK_0000002` there and `rak:0000002` in the Turtle listings above; the two forms denote the same IRI.

3 Data records

3.1 Record types and accounting

The resource distinguishes a field specimen from an extraction, library, sequence run, quality-controlled community profile and derived observation. The resource includes a machine-readable ledger linking

$$\begin{aligned} &\text{site and trip} \rightarrow \text{specimen} \rightarrow \text{extraction} \rightarrow \text{library} \\ &\rightarrow \text{sequence run} \rightarrow \text{QC decision} \rightarrow \text{derived community.} \end{aligned}$$

The versioned ledger contains 2,550 source rows: 2,302 master-sheet rows (2,268 non-control and 34 control rows) plus 248 plant rows. Of these, 2,540 pass the metadata-completeness audit, and all 2,516 non-control rows currently satisfy the knowledge-graph sample-ingest rule. The ten rows that fail the completeness audit are all Trip 1 master-sheet laboratory controls (**Sample_Mastersheet.xlsx**, sheet **Trip1**, rows 578–587): six reagent controls, two PCR controls and two extraction controls. They fail because a control carries no site and no compartment, not because a field record is damaged, and their ledger exclusion reason is **control_requires_explicit_control_model**. They are retained in the ledger with their source identifiers and sheet row numbers. They are not promoted to field specimens; the separate control ABox represents evidence-supported control materials, roles, processing batches, sequence occurrences, assertions and unresolved dispositions. The generated field-specimen graph contains 2,516 sample individuals, 1,647 DNA-extract individuals, 1,242 amplicon libraries and 1,242 FASTQ datasets. The canonical Trips 1–5 feature table contains 1,271 profiles (1,242 unique field identifiers), whereas the ecological analysis retains 1,237 profiles (1,209 unique field identifiers) across 64 numeric sites. Ten of those retained profiles belong to Trip-1-only sites 61–64; the primary repeated-campaign analysis therefore contains 1,227 profiles from sites 1–60. The 29 extra profiles in the canonical feature table arise from 29 normalized identifiers that each occur in two export columns. A count-vector audit found that none of those 29 pairs is identical. After ecological filtering, both profiles remain for 28 of the identifiers. Because the available provenance does not establish whether these are technical reruns, separate aliquots or another export mechanism, the profiles remain distinct and are not silently averaged or deduplicated. An older 1,013-profile table remains in the archive staging tree for provenance, but the package manifest marks it and its companion taxonomy table as legacy artifacts excluded from the canonical dataset. The canonical Trips 1–5 feature table is staged separately and checksum-verified.

The 34 rows not ingested as specimens are all laboratory controls and are referenced by the separate control model rather than forced through the field-specimen shape. The 36 genuine Trip-1-only records from numeric sites 61–64 are ingested through the coordinate-confirmed alias ledger: their source identifiers are preserved while their specimens and visits link to the four existing named site individuals. These rows contribute 11 profiles to the canonical feature table and, after ecological filtering, 10 profiles to the ecological table; the two numbers refer to the two different tables and are not a discrepancy. They remain excluded from repeated-campaign inference because

the sites could not be revisited. After applying the confirmed metadata corrections and aliases and regenerating the sample and DNA modules, eligible-source/sample and expected/generated-DNA counts reconcile exactly. The remaining 26 SRA rows outside the field-specimen lineage are all control-labelled records (18 EB, six **Negative** and two **T.Neg** records) and are represented, where evidence permits, as control sequence occurrences rather than specimen-linked runs. The 34 feature-table profiles absent from the ecological table comprise 24 controls, nine biological-profile identifiers below the declared 1,000-read ecological threshold and one additional **T1Dr1** run with 934 reads, also below that threshold.

3.2 Companion-analysis derived records

The release candidate stages four derived input products used by the companion ecology analyses (Table 3). They are distributed records rather than knowledge-graph ABox modules.

Table 3 Derived companion-analysis inputs in the staged release candidate. Counts refer to the analytical unit identified in the third column.

Staged record	Count	Analytical scope
<code>PMA_ASV_table.tsv</code>	18	Analytical sample columns forming nine treated–untreated aliquot pairs
<code>coverm_profiles.tar.gz</code>	150	Sample-level genome-abundance profiles
<code>eq.emapper.annotations.gz</code>	990	Genomes matched between the annotation table and the CoverM profiles. The count is matched genomes, not annotation rows: the file has one row per annotated protein
<code>measured_function_inputs.tar.gz</code>	6	Exact source tables used by the measured-function analysis

The PMA matrix retains control columns in addition to the 18 paired analytical columns. The measured-function archive includes the genome-derived and PICRUST2 KO inputs, sample mappings, marker records and denominator-audit table used by the reported comparisons. Together, the four products permit downstream statistical re-execution. They do not include the raw shotgun or PMA reads, a full metagenome-assembled-genome catalogue, or the inputs needed to reproduce upstream assembly, binning, dereplication, annotation and profile generation.

3.3 Taxonomy mapping records

The combined taxonomy source contains 354,251 feature assignments in 4,305 distinct seven-rank lineages; 351,472 of those features occur in the canonical 1,271-profile feature table. Coverage is complete in the direction that matters for reuse: every one of

the 351,472 canonical features carries a lineage, and none is unclassified. The remaining 2,779 assignments describe features that have no column entry in the canonical table and no entry in the retired legacy table either, so they contribute no counts to any analysis. The pipeline step that dropped them is not recorded in the exported artifacts, so they are retained as classified-but-uncounted provenance rather than explained away. Expanding each lineage over the seven ranks yields 30,135 mapping-row occurrences but only 8,133 distinct rank-truncated source contexts. Of those occurrences, 20,045 use validated NCBI identifiers, 2,509 use one of 592 unambiguous historical project identifiers, and 7,581 use deterministic contextual identifiers. The latter correspond to 5,030 unique contextual classes. Four historical project identifiers reused across incompatible contexts (36 explicit historical rows) are not retained; all their affected contexts receive lineage-scoped identifiers. No additional project identifier required demotion after the corrected-parent fixed-point audit.

The clean taxonomy module declares the 5,622 local classes (592 retained and 5,030 contextual) plus 2,666 NCBI classes needed for the complete native ancestor closure. Its RDF/XML and Turtle serializations are graph-equivalent at 55,628 triples. The 119 non-identical supplementary species calls comprise 98 calls where the seventh rank was empty and 21 that conflicted with a populated seventh-rank value; all remain in the provenance ledger and none is promoted into the asserted lineage.

The checksum-frozen abundance ABox is 6,722,670,815 bytes and contains 44,528,482 triples in 5,953,690 contiguous subject blocks. It represents 1,236 feature profiles with resolvable FASTQ individuals as 1,236 processing individuals and 17,304 absolute or relative datasets (14 per process). Across seven ranks it records 2,962,918 abundance qualities and 2,962,918 corresponding value individuals, divided evenly between absolute and relative representations (1,481,459 of each quality and value type). The ABox has 1,236 FASTQ bearer subjects and 8,056 taxon bearer subjects. Controls and the 11 Trip-1-only feature-table profiles from sites 61–64, for which no SRA run is recorded, are not silently assigned to taxonomy processes or FASTQ bearers. Control sequence occurrences and unresolved mappings are encoded in the separate control ABox; the Trip-1-only sample and site individuals are encoded in the field modules. Their source observations remain in the release ledger and taxonomy source tables.

3.4 XRF data layers

The XRF distribution contains separate files for the two acquisition workflows and for each processing stage (Table 4):

The current OWL module was generated from all 71 field sessions and all 725 laboratory processed rows, yielding 796 XRF process individuals and 19,763 measurement-value individuals, each with a value-to-quality link. That module and the independently generated XRF audit are synchronized into the archive staging directory with their SHA-256 checksums. The audit reports source, processed, analytical-subset, ecology-joined and KG-ingested counts separately. The canonical input is the 725-record union; regression tests reject the retired 158- and 705-record tables as analytical dependencies. The distribution also includes the canonical 93-row chemical mapping, its pinned PubChem property snapshot, and a row-level ledger recording the resolved

Table 4 Current XRF record inventory. Row counts are independently checked from the processed tables by `scripts/validation/audit_xrf_inventory.py`; provenance and discrepancy fields are generated by `scripts/xrf/audit_xrf_provenance.py`.

Data layer	Rows	Meaning
Field-XRF table	71	Complete in-situ handheld sessions at 58 Trip 5 sites
Laboratory-XRF, Trips 1–4	547	Processed selected-sample records
Laboratory-XRF, Trip 5	178	Processed selected-sample records
Laboratory-XRF, canonical cross-trip input	725	All 547 Trips 1–4 and 178 Trip 5 processed records
Laboratory-XRF, retired Trip 5 table	158	Legacy table retained only for provenance; not used analytically
Laboratory-XRF, retired cross-trip table	705	Legacy 547-plus-158 table; not used analytically

labels, formulae, charges, entity types and audited null dispositions. The supplementary chemical-mapping table (label `tab:chebi_mapping` in the separate supplement) is generated from that ledger and fails closed if any row is not verified.

3.5 Environmental field records

The five immutable field sheets and the original supporting workbook are distributed with a 24-row exact-cell correction ledger. The generated canonical table contains 274 nonblank field records. Its dispositions comprise eight Trip 2 column shifts, 15 workbook-supported campaign-date restorations and one quarantined out-of-range Trip 5 humidity observation. The machine-readable audit records the input hashes, applied ledger entries, record roles, measurement counts, physical ranges and generated-output hashes. The supplementary environmental table and the environmental RDF module consume the same ledger; neither contains the invalid 194% humidity assertion. The table retains all 274 records, while the RDF module links 260 visits to catalogue sites. Fourteen auxiliary/revisit rows with non-catalogue labels remain table-only rather than being assigned by coordinate coincidence. The graph-linked coverage is 260 temperature, 187 pressure and 251 humidity values.

3.6 Archived-soil pH records

The pH distribution is versioned as `EQ-PH-SHARED-v1.0.0`. The frozen source workbook, manifest, version registry and retained predecessor are under `metadata/samples/ph/`. The row-level audit, admitted-measurement table, 29 reconstructed-session table and stable entity registry are under `metadata/ph/`. Together they account for 1,168 source rows: 712 admitted measurements, 356 pending rows, 45 depleted specimens, 36 date quarantines and 19 numeric rows quarantined for incomplete recorded quality control. The source notes identify `S28Sr1` as missed during preparation and `S57Dr1` as depleted; neither receives an imputed value.

The canonical pH module is distributed as graph-equivalent RDF/XML and Turtle files named `ontology/rubalkhali_ph_eq_ph_shared_v1_0_0.owl` and `ontology/rubalkhali_ph_eq_ph_shared_v1_0_0.ttl`. It contains 712 measuring processes, 712 acidity qualities, 712 pH value individuals and 29 sessions. The corresponding source reconciliation, complete ecology sensitivity, predecessor comparison, Shape Map result and output checksums are under `evidence/ph/`. In particular, `evidence/ph/ecology/ph_sample_profile_join.tsv` records the 702 exact specimen-profile joins, `evidence/ph/ecology/ph_group_analysis_table.tsv` records the 562 site-campaign-position means, `evidence/ph/ecology/ph_geographic_adjustment.tsv` records the same-cohort spatial comparison, and `evidence/ph/ecology/ph_influence_sensitivity.tsv` records the singleton maximum and whole-site exclusions. The composition analysis uses the 558 groups reported in `evidence/ph/ecology/ph_composition_models.tsv`. Of the predecessor's 653 admitted observations, all 653 are unchanged and retain their IRIs; the shared version adds 59 admitted observations.

3.7 Knowledge-graph modules

The distribution is organized into reusable modules for the base terminology, sites, specimens, environmental and climate measurements, XRF, DNA extractions, sequence records, sequence QC and taxonomic observations, plus separate laboratory-control and archived-soil pH modules. The staging package contains the generated modules, the source metadata included in the ledger, RDF generators, ShEx schemas, validation scripts, and the generated release and XRF evidence.

3.8 Deposited files, formats and checksums

The deposit is a single directory tree. Every payload file is declared; the package builder adds `PRE_RELEASE_MANIFEST.tsv` itself explicitly rather than making the manifest recursively list its own checksum. Table 5 gives the top-level layout, the format of each part and what a reuser can do with it. Paths in this section and in every listing are relative to the root of that tree.

Table 5: Layout of the deposited package. Every file under these paths appears in `PRE_RELEASE_MANIFEST.tsv` with its byte size, SHA-256 checksum, category, release status and licence gate.

Path	Format	Content and reuse
<code>ontology/</code>	RDF/XML, Turtle	Terminology and the ten ABox modules; loadable into any RDF store. The taxonomy ABox is a single 6.72 GB Turtle file
<code>ontology-src/</code>	Turtle	Modular source terminology used to build the knowledge graph
<code>void.ttl</code>	Turtle	Candidate dataset description without unresolved licence or live-service assertions

Path	Format	Content and reuse
metadata/DATA_DICTONARY.tsv	TSV	Machine-readable field names, data types, units, missing-value conventions and reuse definitions for curated canonical tabular records; raw instrument exports retain their source headers
shex/	ShEx	One shape per entity family; the structural contract the modules satisfy
sparql/	SPARQL, Markdown	The validated competency query and query/schema notes
metadata/samples/	XLSX, TSV	Source master sheet, plant sheet, correction and alias ledgers
metadata/samples/ph/	XLSX, JSON, TSV	Frozen pH source versions, manifests, registry and version policy
metadata/ph/	TSV	Complete pH row audit, 712 admitted observations, 29 sessions and stable entity registry
metadata/controls/	TSV, JSON	Normalized control entity, alias, role, process, sequence-occurrence, composition, assertion and disposition records
metadata/samplesheets/	TSV	Five immutable per-campaign field sheets
metadata/environmental/	TSV	274 range-checked environmental records derived from those sheets
metadata/climate/	TSV	Open-Meteo daily and monthly tables, the fetch-key ledger with each key's resolved site IRI, and the site-resolved projections
metadata/geodata/	TSV, JSON	Per-campaign coordinates, altitudes and the site-to-campaign map
metadata/geochemistry/, metadata/xrf/	TSV, XLSX	Field log, processed laboratory tables and the raw instrument exports
metadata/taxonomy/	TSV, FASTA	Canonical feature table, taxonomy table and ASV sequences, plus the legacy tables marked legacy-excluded
metadata/protocols/	Markdown	Extraction, library-preparation, sequencing and measurement protocols with YAML front matter
metadata/QC_reads/	TXT	MultiQC per-run FastQC statistics
metadata/amplicon/	CSV, TSV, config, Markdown	Portable 1,268-row run sheet, accessions, canonical command and resources
metadata/functional/	Markdown	PICRUSt2 execution and reference manifest with stated input limits

Path	Format	Content and reuse
metadata/ sra-submissions/	TSV	Two source submission sheets from which run-level metadata are reconciled
metadata/ metagenome/, metadata/ relic-dna/ evidence/	TAR/GZ, TSV	The four companion-analysis inputs of Table 3
evidence/ph/	JSON, TSV, MD	Machine-readable audit outputs and SHA256SUMS files for the release, XRF, environmental, taxonomy, controls, competency-query and companion-input gates
evidence/ ecology-canonical/	TSV, JSON	pH source reconciliation, ecology analysis, cross-version comparison, ShEx result and checksums
evidence/ xrf-community/, evidence/ primer-identity/ scripts/, config/	TSV, MD	Exact baseline cache and canonical verdict files consumed by the package-only control-sensitivity replay
workflow/, environment/	TSV, MD	Laboratory-XRF axis inputs and the raw-read primer-identity audit, including its source-path ledger
tests/	Groovy, Python, YAML	The generators, validators and code tables that produce the modules
manage.sh, deploy_onto.sh	Nextflow, shell, YAML, TXT	Orchestration, task wrappers, workflow tests, a pinned Conda recipe and a hash-locked Python environment
	Python, Groovy, Turtle	Unit tests and positive/negative structural fixtures distributed with the candidate
	shell	Operational graph-management and loading references; deployment is not required for local reuse

Tabular records are tab-separated with a header row and no index column, except for the explicitly named portable CSV samplesheet; RDF modules use the prefixes of Table 2. `metadata/DATA_DICTIONARY.tsv` records the field or field pattern, data type, unit, missing-value convention and definition for curated canonical tabular records; raw instrument exports retain their source headers. Quantities carry units only where a unit is documented at the source. Environmental source tables retain degrees Celsius, hectopascals (millibars) and percent relative humidity. In `ontology/rubalkhali_measurements.owl`, pressure magnitudes are multiplied by 100 and asserted in pascals (`obo:UO_0000110`); climate records use degrees Celsius and millimetres; DNA concentrations are in nanograms per microlitre. Archived-soil

pH values use UO:0000196 and are bound to the specified CaCl_2 protocol. XRF concentrations carry no documented unit or unit assertion and must not be treated as percentages.

The package-level manifest `PRE_RELEASE_MANIFEST.tsv` is exhaustive over payload files: it has one row for each candidate payload file, and the builder fails if a declared file is absent or a present file is unclassified, so the manifest cannot drift from the tree. Each row carries the relative path, a category, a release status (`canonical-candidate`, `audit-evidence`, `legacy-excluded`, `derived-review-required` or `operational-reference`), a one-line record scope, the byte size, the SHA-256 checksum, and a licence status. Licences are recorded as open gates rather than asserted: project-produced files await the depositor's licence selection, while redistributed content derived from NCBI Taxonomy, the imported OBO ontologies or the SILVA reference database retains a separate third-party licence gate tied to the pinned upstream release. Legacy taxonomy and XRF tables remain available for provenance but are explicitly excluded from canonical analytical dependencies.

Scope boundaries of the deposit, stated once: it contains no raw sequencing reads (these are in the sequence archives, see Section 6), no metagenome-assembled-genome catalogue, and no assembly or binning intermediates. The canonical feature table contains 24 extraction-blank community profiles. The package records the author-confirmed product, stage and assay facts in `evidence/controls/control_ground_truth.tsv`; normalized control registries and process tables in `metadata/controls/`; the generated SIO-patterned control graph in `ontology/rubalkhali_controls.ttl`; the candidate-ASV, expected-recovery and exact-ASV-overlap records in `evidence/control-audit/`; and all 25 before/after headline metrics across 11 claims in `evidence/control-sensitivity/headline_result_sensitivity.tsv`. The control graph contains 489 entities and 3,800 triples, including 51 material-specific role individuals and 47 sequence occurrences. It represents a control as a material bearing a role realized in a laboratory process, links extraction processes to batches rather than trips, and expresses uncertain identities or mappings as evidence-bearing dispositions instead of ground truth.

The downloadable reviewer archive omits three payloads above its 256-MiB threshold and lists them with byte size and SHA-256 checksum in its separate bulk manifest: `metadata/metagenome/eq.emapper.annotations.gz`, `metadata/taxonomy/feature-table-trips1-5.tsv`, and `ontology/rubalkhali_taxonomy_abox.ttl`. A complete deposit must supply those three byte-identical files; their omission is only a reviewer-download packaging choice.

4 Technical validation

Technical validation is organized around the evidential question each test can answer. Source-to-graph reconciliation checks accounting; XRF replicate and cross-workflow analyses check measurement behaviour; ShEx checks graph structure; bounded ELK classification checks a stated OWL fragment; and executable SPARQL checks a specific retrieval path. Passing one layer is not reported as validation of another.

4.1 Source reconciliation and measurement validation

The executed evidence workflow produced a row-level sample ledger and reconciliation summaries for specimens, extracts, libraries, runs and community profiles. The 2,516 eligible source rows reconcile exactly to 2,516 generated sample individuals; the 1,647 extraction-eligible rows reconcile to 1,647 DNA extracts; and 1,242 resolved sequence rows reconcile to 1,242 amplicon libraries and 1,242 FASTQ datasets. The canonical feature table contains 1,271 profiles, while ecological filtering retains 1,237: 1,227 from the sites 1–60 primary frame and 10 from the Trip-1-only sites 61–64. The profile-identity audit also reproduced 29 pairs of export columns that collapse to the same field identifier in the canonical table, found zero identical count-vector pairs and verified that 28 pairs remain after ecological filtering. This audit prevents the repeated identifiers from being mistaken for identical copied columns; it cannot recover their undocumented upstream laboratory or export relationship.

The site-alias gate independently matches each Trip 1 numeric identifier 61–64 to a named catalogue individual by exact coordinates and rejects coordinate or IRI disagreement. It accounts for all 36 affected source rows. The regenerated environmental module also resolves all four coordinate-matched named Trip 1 visits, and the sample graph links every affected specimen to both its canonical site and the corresponding visit without changing the numeric source identifier or repeated-campaign exclusion. An additional cross-module gate checks all 1,004 sample-process `sio:000068` links against the regenerated environmental module. All 252 distinct visit targets resolve, and every linked visit targets the same catalogue site as the sample process through `sio:000291`. This gate prevents a sample module generated against an earlier visit numbering from passing validation merely because it is internally well formed.

The environmental-metadata gate reproduced 274 nonblank source records and verified all 24 exact-cell dispositions against the corresponding raw cells. It retained the workbook-supported Trip 3 site 21 humidity of 31.321%, restored 15 appended observations to their original March 2023 dates, and confirmed that the resulting Trip 3 expedition contains 65 February 2024 records ending on 21 February. All retained field temperatures, pressures and relative humidities fell within their declared physical ranges. The unresolved Trip 5 site 40 value of 194% was preserved in the ledger but absent from both the curated table and the regenerated environmental RDF module. Regression tests also ensure that the Trip 2 termination note cannot leak into a measurement column. The regenerated module links 260 of the 274 rows to unambiguous catalogue visits and contains 260 temperatures, 187 pressures and 251 relative-humidity values. Fourteen auxiliary or revisit labels remain table-only because coordinate coincidence does not establish catalogue identity; exact catalogue label 19.5 is the sole auxiliary record linked to a visit.

The high-risk XRF inventory was independently counted by `scripts/validation/audit_xrf_inventory.py` and reconciled by `scripts/xrf/audit_xrf_provenance.py`. The audit reproduced 13,390 reported Trips 1–4 cells under the maximum-positive rule and 4,293 reported Trip 5 cells under the last-reported rule, with no mismatched reported cells. It also verified that the canonical laboratory input comprises all 725 processed records and that the retired 158- and 705-row artifacts are absent from analytical dependencies. The release-evidence and XRF regression

suites comprised automated source, aggregation and manuscript-consistency tests, all of which passed for the staged modules.

The independent chemical-identifier gate `scripts/xrf/audit_xrf_chemical_mapping.py` checked all 93 instrument channels against the checksum-recorded, pinned ChEBI v247 OWL file and a checksummed PubChem snapshot retrieved on 23 July 2026. All rows passed. The resulting ledger contains 79 supported ChEBI and 91 supported PubChem cross-references; audited nulls are retained for LE, unsupported ChEBI oxides and the PrO_2 channel rather than being replaced by lexical matches. The pinned ChEBI v247 and NCBITaxon OWL source files are identified by checksum but are not redistributed; re-executing the corresponding identifier gates requires reacquiring those exact upstream versions.

The field/laboratory diagnostic is a site-level comparison, not a same-aliquot validation. Missing machine-readable unit, detection-limit and non-detect semantics limit absolute comparisons and interpretation of zero values; structural graph validation does not resolve those measurement limitations.

The pH reconciliation independently accounts for all 1,168 workbook rows and matches every admitted row to exactly one existing specimen. It admits 712 values grouped into 29 reconstructed sessions and reproduces the five dispositions reported in Data Records: 712 admitted, 356 pending, 45 depleted, 36 date-quarantined and 19 quality-control-quarantined rows. Source size and checksum are verified against the frozen version manifest. The cross-version gate reports 653 unchanged predecessor observations, 59 additions, and no absent, revised-value or revised-metadata observation. Because measurement date contributes to the stable entity key, a corrected date would be reported as one absent and one added observation rather than as revised metadata; either outcome fails the no-change gate. This validates version membership and source accounting, not whether an unavailable specimen would have followed the same pH distribution.

4.2 Structural validation

We used Shape Expressions (ShEx) [45] for closed-world checks of required properties, cardinalities and datatypes. The current site, specimen, environmental-measurement, DNA, XRF, sequence-record and QC modules passed their assigned shapes. The pH gate additionally validates every one of the 712 measurement processes, values and acidity qualities, together with all 29 sessions, against `shex/ph_measurements.shex`. The deposited `evidence/ph/ph_validation.shexmap` enumerates all 2,167 node-shape pairs. A minimal negative graph with UO:0000196 removed from one pH value, its shape map and the nonconformance log are distributed alongside it under `evidence/ph/`. The canonical Turtle and RDF/XML files are graph-equivalent and the root knowledge graph imports their shared version IRI. This establishes conformance of the tested nodes to the declared schemas, not source accuracy, completeness, absence of contamination or biological plausibility.

The multi-gigabyte taxonomy ABox remains outside the memory-bounded full-ShEx run. It is instead generated in an isolated workflow task, checksum-identified, parsed in full by an independent Turtle parser and scanned by a memory-bounded

structural validator. That scan requires every generated classification process, absolute and relative dataset, abundance quality and abundance value to have the expected types, directions and cardinalities; it also reconciles the complete generated identifier ranges and rejects a truncated file. The resulting evidence is full-file syntax and generator-specific structural validation, not a claim that the taxonomy ABox passed the unexecuted ShEx shapes. The tractable modules' ShEx suite tests both conformance and rejection. Three positive fixtures (a georeferenced sampling site, a canonical climate measurement and a laboratory-XRF process with a specimen input) must conform, and four negative fixtures must be rejected by the released shapes: a sampling site without `geo:asWKT`, a measurement payload whose value carries an invalid datatype, a measurement in which the quality rather than the value is the subject of `sio:000215`, and an XRF process with neither an input nor a target. Each negative fixture differs from its positive counterpart in exactly one respect, so a passing negative test identifies the constraint that caught it. A further positive fixture exercises the complete laboratory-control pattern, and two single-defect control fixtures reject an extraction blank realized at the wrong processing stage and an expected-taxon assertion whose composition specification lacks source provenance. The suite is `tests/shex_negatives/run_tests.groovy`; all ten fixtures behave as declared. The coverage is deliberate rather than exhaustive: it targets the four defect classes that the modules are structurally able to admit, and it does not enumerate every constraint in every shape. A separate query gate, `scripts/validation/validate_site_biome_completeness.py`, supplies the closed-world check that no shape can express, requiring each of the 70 sampling-site individuals to carry a biome and an environmental-feature restriction with ENVO fillers. In the final workstation run, the structural scan read all 6,722,670,815 bytes and 50,482,175 lines of the file whose SHA-256 is `d78be36a0fb3f91a28bec4c6489ca1a34d97c576e10cd2c92fc59e4dfd0eb808`. A reader should expect that digest and that byte count for the deposited `ontology/rubalkhali_taxonomy_abox.ttl`; the staged deposit copy and the generated module both carry it. The scan accounted for 44,528,482 triples and 5,953,690 subject blocks with zero structural violations or warnings. Raptor independently parsed the complete Turtle file and reported the same 44,528,482 triples with no diagnostic warning or error. The two passes also reconciled all 1,236 processes, 17,304 datasets, 2,962,918 qualities and 2,962,918 values.

Before ABox generation, the complete taxonomy mapping passed coverage, rank, exact-label-or-synonym, NCBI-ancestry, contextual-IRI-collision and shared-prefix-consistency checks. The builder quarantined the single historical source-header artifact, covered all 4,305 canonical lineages with 30,135 rank rows and reported zero missing lineages. It also verified graph equivalence of the 55,628-triple Turtle and RDF/XML taxonomy modules and rejected identity axioms or any reparenting of an NCBI class beneath a local class. These checks validate the stated mapping policy and serialization contract; they do not turn a contextual class into an externally identified taxon.

4.3 Formal-semantic validation

ELK classification is limited to OWL 2 EL [46]. The gate runs ELK 0.4.3 through the OWL API 4.5.26. A clean 1 August 2026 workstation rerun on the current tractable modules loaded 781,293 axioms from 10 distinct ontology IRIs using local module and ontology caches; sibling serializations sharing an ontology IRI were deduplicated. ELK reported the resulting merged ontology as consistent. The run excluded the multi-gigabyte taxonomy ABox and omitted the retired pre-audit project-taxonomy and NCBI-subset modules; the generated canonical ecosystem and laboratory-control modules were included. It did not precompute the class hierarchy. A separate label gate merged 782,229 triples from 11 current files and confirmed exactly one label for each of 104,697 labelled RAK_ subjects. All assigned ShEx checks and the IRI-registry gate also passed, as did the control-specific invariant validator. The exact log, input hashes and validator hashes are archived under `evidence/semantic-validation/`.

A separate 1 August 2026 control-semantic run validated every shaped node in the 3,800-triple control module, not a sample of nodes. It confirmed 489 registered entities, 18 expected-taxon assertions for 18 manufacturer composition rows, graph equivalence of the Turtle and RDF/XML serializations, the four declared SIO inverse pairs, the ShEx-required input, target and output links, generator determinism and merged bounded ELK consistency. Its machine-readable record, log and hashes are archived in the same evidence directory.

This result is not a full OWL 2 DL consistency claim. ELK is incomplete for universal restrictions, inverse properties and other constructs used by imported modules, and the excluded ABoxes were not covered. Likewise, the selected forward-rule program has not been demonstrated to reach a fixed point and is not described as complete OWL-Horst closure [54].

4.4 Worked-example validation

The worked Turtle patterns of Section 2.12 are the part of this descriptor a reuser is most likely to copy, so they are checked mechanically rather than maintained by hand. `scripts/validation/verify_manuscript_listings.py` reads the prefix table out of the manuscript, parses each Turtle listing, and resolves every resulting triple against the staged modules, including the `owl:Restriction` structures that the sampling-site pattern depends on and the multi-gigabyte taxonomy ABox. It additionally requires every prefixed IRI named in the section prose to exist in the graph, which is what catches a malformed term or a property that was never minted. The current listing gate resolves all 99 printed triples across the nine Turtle listings and all 55 prose IRIs. A drifted listing therefore fails a test instead of reaching a reader as an unusable query pattern.

4.5 Executable field-XRF query validation

Listing 5 follows the corrected field-XRF path. The process targets the recorded site with `sio:000291`; it does not claim a physical specimen input. The produced value points to its quality with `sio:000215`. The exact query is archived as `sparql/field_xrf_site10.rq`. The workflow executed `scripts/validation/`

`validate_competency_query.py` against the checksummed base, site and XRF modules and required the fixed expected cardinality of 46 analyte-value rows. It returned exactly 46 rows for Site 10: 24 from field process Test 5847 and 22 from Test 5848. The validator also required exactly two process labels and no site label other than Site 10, and wrote the result table, input checksums and machine-readable validation record under `evidence/competency-query/`.

Listing 5 Retrieve field-XRF analyte values for a selected site.

```
PREFIX rak: <https://rubalkhali.science/kb/RAK_>
PREFIX sio: <http://semanticscience.org/resource/SIO_>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>

SELECT ?processLabel ?siteLabel ?analyte ?concentration
WHERE {
  ?process a rak:0000025 ;
           rdfs:label ?processLabel ;
           sio:000291 ?site ;
           sio:000229 ?value .
  FILTER(CONTAINS(LCASE(STR(?processLabel)), "field xrf"))
  ?site rdfs:label ?siteLabel .
  FILTER(?siteLabel = "Site 10")
  ?value sio:000215 ?quality ;
         rak:2000012 ?concentration .
  ?quality a ?qualityClass .
  ?qualityClass rdfs:label ?analyte .
  FILTER(?qualityClass != rak:0000032)
}
ORDER BY ?processLabel ?analyte
```

This establishes only the archived, bounded field-XRF retrieval and its 46-row cardinality. No broader competency-query pass rate is claimed: several historical queries lacked fixed expected cardinalities, and the unbounded cross-run taxonomy query exceeded its development timeout. Those queries are not used as technical-validation evidence in this descriptor.

5 Usage notes

The modular graph can be loaded locally or into an RDF store and queried with SPARQL 1.1. A service and portal are intended for <https://rubalkhali.science/sparql> and <https://rubalkhali.science/>, respectively, but anonymous access to these deployments was not operationally verified at manuscript revision. They are therefore not evidence for any result reported here and are not required to use the archived modules.

The intended cross-domain query path joins laboratory calcium values to community profiles through explicit trip–site–compartment and specimen provenance. Users can then compute diversity measures such as Shannon’s index from the retrieved abundances. This cross-domain path remains a design use case rather than a validated competency-query result. By contrast, the bounded field-XRF retrieval in Section 4 is workflow-validated against the three checksum-identified modules at a fixed cardinality of 46 rows. That single result does not establish a pass rate for the broader historical query set.

The modular OWL architecture permits extending the knowledge graph with new data types, such as metabolomics or remote sensing data. The release life-cycle, comprising data preparation, graph generation, validation, evidence-bundle creation and manuscript-derived statistics, is orchestrated from a single Nextflow entry point (`main.nf`). Lower-level scripts remain implementation details invoked by that workflow.

5.1 Design rationale

We adopted the SemanticScience Integrated Ontology (SIO) as the upper-level framework because it offers a small set of relations (`hasInput/hasOutput`, `hasTarget`, `hasMember`, `isMeasurementValueOf`, `hasMeasurementValue`, `hasValue`) that generalize across sampling, measurement, and provenance patterns. Sample collection, laboratory XRF, DNA extraction, sequencing and taxonomic classification use the `Process--hasInput/hasOutput--Object` idiom. Field XRF instead targets the recorded site because its source log lacks a physical specimen identifier. Quantitative results are uniformly represented as reciprocal `measurement value--quality` pairs. A unit is asserted only where the source documents one. XRF concentration values have no unit assertion because the source records do not document one. The uniform pattern is what allows one query shape to traverse a climate observation, a sampling site and a taxon abundance without per-domain join logic.

Ingest validation uses ShEx because the ETL is batch-oriented and the validator runs in the same Groovy/Jena toolchain that produces the RDF. SHACL would provide equivalent checks at load time for a reuser who prefers in-database validation. The modules load into any SPARQL 1.1 store; the reference deployment uses Virtuoso.

Full OWL 2 DL reasoning over the combined TBox and ABox is not currently feasible with HermiT on the available hardware. ELK [46] therefore classifies the OWL 2 EL modules, while complete reasoning is restricted to the TBox and bounded representative modules. For common query paths, a selected forward-rule program materializes only its documented RDFS/OWL rule families. It is run to three scheduled passes, including one transitive pass, but has not been demonstrated to reach a fixed point. It is therefore not described as complete OWL-Horst closure [54].

5.2 Limitations and outstanding challenges

Several limitations remain. ELK does not validate arbitrary OWL 2 DL axioms, and the selected materializer does not implement every OWL entailment. Taxonomy correspondences remain uncertain outside the direct SILVA-to-NCBI mapping; unreviewed GTDB and iNaturalist candidates are excluded rather than asserted as equivalences. Archived-soil pH is available for 712 admitted specimens under the specified 1:2.5 CaCl₂ protocol, but coverage is incomplete and non-random and should not be interpreted as an in situ field map. Electrical conductivity remains unavailable. The XRF sources lack machine-readable unit and non-detect semantics, and Trips 1–4 and Trip 5 use disclosed but different repeat-formula rules. The canonical analysis includes all 725 processed laboratory records; the former 158- and 705-row tables are retired. Field-XRF and laboratory-XRF values cannot be assumed interchangeable, and their

current comparison is site-level rather than same-aliquot. SIO, ChEBI, ENVO and NCBI Taxonomy evolve independently, so each dataset version records its imported artifacts and mapping decisions.

Three further limits bear directly on what the package can be used for. First, the assay-aware screen identifies 351 of 351,472 ASVs using 17 extraction-day blanks with batch links to Trip 5 profiles. Removing these candidates only from the 217 co-extracted Trip 5 biological profiles removes 2.19% of reads in aggregate (median 0.403%; maximum 56.6%). Shannon diversity before and after filtering has Spearman $\rho = 0.995$, no profile falls below 25,000 reads, and all 25 tracked metrics across 11 claims remain stable; no claim verdict changes. Only two of seven positive-labelled profiles are mock-dominated under the declared 95% descriptive threshold. Their library-to-product mappings remain provisional trip-design hypotheses. Exact mock-associated ASV overlap elsewhere is descriptive rather than proof of index hopping or cross-contamination [43]. The unfiltered table remains canonical. Inference is bounded because the design provides only one blank per extraction batch, concentration data are absent, Trip 4 has no positive control and has an unsequenced blank affecting all or part of 23 sites, and Trip 5 has no 16S positive control. Three sequenced PCR-blank libraries and one Trip 4 PCR-blank record are represented, but the complete PCR-blank-to-batch mapping awaits laboratory confirmation. The sterile-bag inventory remains unavailable. A reuser should therefore use the control ABox and audit tables to select compatible controls by assay, stage and batch, and should not treat contamination as eliminated. Second, the manifest is exhaustive over what is deposited, but what is deposited is bounded. The package includes a portable control-sensitivity replay, the paired PMA table, the measured-function archive, and the CoverM and eggNOG inputs used by the encoded-function test. These support the named downstream endpoints; they do not reproduce raw-read processing, assembly, binning, dereplication, annotation or profile generation. Third, per-file licences are recorded as open gates rather than asserted values; the deposit licence and the upstream licences of redistributed taxonomy and ontology content must be resolved at deposition before any file here is redistributed further.

5.3 Sustainability and updates

The versioning policy treats each deposited dataset as immutable once it is archived under a persistent identifier; the staged candidate described here is not yet archived and is therefore still mutable. Subsequent expeditions or upstream-ontology updates receive a new version rather than silently modifying the cited files. The manifest records file checksums, and changes to mappings and imports are tracked between versions.

6 Data availability

Three project accessions appear in the project records. They were resolved against the live European Nucleotide Archive on 28 July 2026; the retrieval, including the verbatim batch responses, the exact query strings and their checksums, is archived at

evidence/accessions/. The accessions are not reconciled, and none of them is currently a usable route to the data described here.

- PRJEB106069 is the project against which all 1,268 submitted amplicon run rows are registered in the submission sheets. It returns no public record.
- PRJEB104209 is a public umbrella study, “The genome of the Rub’ al Khali (Empty Quarter)” (secondary accession ERP185514, first public 23 November 2025), but it carries no read runs of its own.
- PRJNA1065643 is public and holds 85 amplicon read runs on Illumina MiSeq, first public 1 July 2024. It is an earlier, partial deposition on a different instrument; 85 runs are not the five-expedition release described here, and its existence is not evidence for that release.

Every run accession the project holds was then checked individually. The 1,268 submitted run rows comprise 1,242 field-specimen-linked rows represented in the sequence module and 26 control-labelled rows represented separately; they cover 1,238 distinct submitted identifiers because 30 identifiers have a second library. *None resolves to a public record.* The `rdfs:seeAlso` accession links in the released graph therefore do not reach anything a reader can retrieve, and no sample-level or run-level mapping between the released records and a public archive record can be asserted. Establishing that mapping requires PRJEB106069 and its runs to be released; until then this remains an open availability blocker rather than a resolved statement. The staged release candidate contains the derived companion-analysis inputs described in Table 3: nine paired PMA aliquots represented by 18 analytical columns, 150 CoverM profiles, eggNOG annotations for the 990-genome analytical subset and the six-table measured-function input archive. The PMA and measured-function products directly support their reported endpoint analyses; the encoded-function null additionally uses the bulk-listed eggNOG file, and the control sensitivity has its own portable replay script.

They do not include the underlying raw shotgun or PMA reads or the full metagenome-assembled-genome catalogue, and they do not make upstream assembly, binning, dereplication, annotation or CoverM generation reproducible. At the time of this manuscript revision, the staged dataset and graph had not yet received an immutable public DOI, and anonymous resolution and download of all project, sample and run accessions had not been verified. Public deposition of the checksummed package and underlying sequence resources, a frozen source commit and verified accessions remain external availability blockers for submission.

7 Code availability

The manuscript, RDF generators, source metadata and reproducibility workflow are maintained in the dedicated repository at <https://github.com/bio-ontology-research-group/empty-quarter-data-paper>. The release candidate contains the ShEx shapes, validation scripts and fixtures, query examples, the `main.nf` entry point, execution profiles, task wrappers, workflow tests, a pinned Conda recipe and a hash-locked Python environment. The broader portal development tree remains

at <https://github.com/bio-ontology-research-group/empty-quarter>. Anonymous public access to the dedicated repository was not verified at manuscript revision. A frozen commit and anonymously accessible source archive are required for submission.

References

- [1] Mandaville, J.P.: Plant life in the rub' al-khali (the empty quarter), south-central arabia. *Proceedings of the Royal Society of Edinburgh. Section B. Biological Sciences* **89**, 147–157 (1986) <https://doi.org/10.1017/s0269727000008988>
- [2] Makhallanyane, T.P., Valverde, A., Gunnigle, E., Frossard, A., Ramond, J.-B., Cowan, D.A.: Microbial ecology of hot desert edaphic systems. *FEMS Microbiology Reviews* **39**(2), 203–221 (2015)
- [3] Fierer, N.: Embracing the unknown: disentangling the complexities of the soil microbiome. *Nature Reviews Microbiology* **15**(10), 579–590 (2017)
- [4] Yasir, M., Azhar, E.I., Khan, I., Bibi, F., Baabdullah, R., Al-Zahrani, I.A., Al-Ghamdi, A.K.: Composition of soil microbiome along elevation gradients in southwestern highlands of Saudi Arabia. *BMC Microbiology* **15**(1), 65 (2015) <https://doi.org/10.1186/s12866-015-0398-4>
- [5] Aslam, S.N., Dumbrell, A.J., Sabir, J.S., Mutwakil, M.H.Z., Baeshen, M.M.N., Abo-Aba, S.E.M., Clark, D.R., Yates, S.A., Baeshen, N.A., Underwood, G.J.C., McGenity, T.J.: Soil compartment is a major determinant of the impact of simulated rainfall on desert microbiota. *Environmental Microbiology* **18**(12), 5048–5062 (2016) <https://doi.org/10.1111/1462-2920.13474>
- [6] Khan, M.A., Khan, S.T.: Microbial communities and their predictive functional profiles in the arid soil of Saudi Arabia. *SOIL* **6**, 513–521 (2020) <https://doi.org/10.5194/soil-6-513-2020>
- [7] Osman, J.R., Zélicourt, A., Bisseling, T., Geurts, R., Hirt, H., DuBow, M.S.: Bacterial rhizosphere biodiversity from several pioneer desert sand plants near Jizan, saudi arabia. *The Open Conference Proceedings Journal* **7**(Suppl. 1: M7), 70–79 (2016) <https://doi.org/10.2174/2210289201607020070>
- [8] Eida, A.A., Ziegler, M., Lafi, F.F., Michell, C.T., Voolstra, C.R., Hirt, H., Saad, M.M.: Desert plant bacteria reveal host influence and beneficial plant growth properties. *PLOS ONE* **13**(12), 0208223 (2018) <https://doi.org/10.1371/journal.pone.0208223>
- [9] Maurice, K., Laurent-Webb, L., Dehail, A., Bourceret, A., Boivin, S., Boukcim, H., Selosse, M.-A., Ducousso, M.: Fertility islands, keys to the establishment of plant and microbial diversity in a highly alkaline hot desert. *Journal of Arid Environments* **219**, 105074 (2023) <https://doi.org/10.1016/j.jaridenv.2023.105074>

- [10] Maurice, K., Laurent-Webb, L., Bourceret, A., Boivin, S., Boukcim, H., Selosse, M.-A., Ducouso, M.: Networking the desert plant microbiome, bacterial and fungal symbionts structure and assortativity in co-occurrence networks. *Environmental Microbiome* **19**, 65 (2024) <https://doi.org/10.1186/s40793-024-00610-4>
- [11] Al-Awthan, Y.S., Mir, R., Alatawi, F.A., Alatawi, A.S., Almutairi, F.M., Khafaga, T., Shohdi, W.M., Fakhry, A.M., Alharbi, B.M.: Metagenome analysis identified novel microbial diversity of sandy soils surrounded by natural lakes and artificial water points in King Salman Bin Abdulaziz Royal Natural Reserve, Saudi Arabia. *Life* **14**(12), 1692 (2024) <https://doi.org/10.3390/life14121692>
- [12] Alhasani, F.M., Rajab, M.H., Sulieman, A.M.E., Alanazi, N.A., Aljabry, N., Alwahibi, M.S., Mammdouh, A.: Diversity and ecological drivers of rhizospheric and endophytic microbiomes of desert plants in King Salman Reserve, Saudi Arabia. *Journal of King Saud University – Science* **38**, 16952025 (2026) https://doi.org/10.25259/JKSUS_1695_2025
- [13] Gopi, J., Joydas, T.V., Jayachandran, P.R., Manokaran, S., Mohandas, S.P., Raju, S.C., Hussain, S.A., Abouelresh, M.O., Dixit, S., Rafeeq, M., Asharaf, M., Manikandan, K.P., Hala, S., Pulido, B.: eDNA metabarcoding reveals halophilic and halotolerant bacterial assemblages in the salt flats of the northern Red Sea coast. *Ecological Genetics and Genomics* **38**, 100448 (2026) <https://doi.org/10.1016/j.egg.2026.100448>
- [14] Moussa, T.A.A., Al-Zahrani, H.S., Almaghrabi, O.A., Abdelmoneim, T.S., Fuller, M.P.: Comparative metagenomics approaches to characterize the soil fungal communities of western coastal region, Saudi Arabia. *PLOS ONE* **12**(9), 0185096 (2017) <https://doi.org/10.1371/journal.pone.0185096>
- [15] Murgia, M., Fiamma, M., Barac, A., Deligios, M., Mazzarello, V., Paglietti, B., Cappuccinelli, P., Al-Qahtani, A., Squartini, A., Rubino, S., Al-Ahdal, M.N.: Biodiversity of fungi in hot desert sands. *MicrobiologyOpen* **8**(1), 00595 (2019) <https://doi.org/10.1002/mbo3.595>
- [16] Mani, I., Mikryukov, V., Alkahtani, S., Tedersoo, L.: Desert mycobiome of Saudi Arabia is driven by vegetation patterns. *MycKeys* **130**, 71–100 (2026) <https://doi.org/10.3897/mycokeys.130.176937>
- [17] Blanco-Sacristán, J., Nelson, C., Corrochano-Monsalve, M., Maestre, F.T.: Biological soil crusts in the Arabian Peninsula: Ecological functions, current knowledge and research gaps. *Cambridge Prisms: Drylands* **2**, 13 (2025) <https://doi.org/10.1017/dry.2025.10007>
- [18] McKay, C.P., Rask, J.C., Detweiler, A.M., Bebout, B.M., Everroad, R.C., Lee, J.Z., Chanton, J.P., Mayer, M.H., Caraballo, A.A.L., Kapili, B., Al-Awar, M., Al-Farraj, A.: An unusual inverted saline microbial mat community in an interdune sabkha in the Rub' al-Khali (the Empty Quarter), United Arab Emirates. *PLOS*

- [19] Mousa, W.K., Abu-Izneid, T., Salah-Tantawy, A.: High-throughput sequencing reveals the structure and metabolic resilience of desert microbiome confronting climate change. *Frontiers in Plant Science* **15**, 1294173 (2024) <https://doi.org/10.3389/fpls.2024.1294173>
- [20] Wilkinson, M.D., Dumontier, M., Aalbersberg, I.J., Appleton, G., Axton, M., Baak, A., Blomberg, N., Boiten, J.-W., Silva Santos, L.B., Bourne, P.E., *et al.*: The FAIR Guiding Principles for scientific data management and stewardship. *Scientific Data* **3**(1), 160018 (2016)
- [21] Yilmaz, P., Kottmann, R., Field, D., Knight, R., Cole, J.R., Amaral-Zettler, L., Gilbert, J.A., *et al.*: Minimum information about a marker gene sequence (MIMARKS) and minimum information about any (x) sequence (MIXS) specifications. *Nature Biotechnology* **29**(5), 415–420 (2011) <https://doi.org/10.1038/nbt.1823>
- [22] Kelliher, J.M., Mirzayi, C., Bordenstein, S.R., Oliver, A., Kellogg, C.A., Hatcher, E.L., Berg, M., Baldrian, P., Aljumaah, M., *et al.*: STREAMS guidelines: standards for technical reporting in environmental and host-associated microbiome studies. *Nature Microbiology* **10**, 3059–3068 (2025) <https://doi.org/10.1038/s41564-025-02186-2>
- [23] National Microbiome Data Collaborative: National Microbiome Data Collaborative: Data Standards. <https://microbiomedata.org/data-standards/>. Accessed 23 July 2026 (2026)
- [24] Lebo, T., Sahoo, S., McGuinness, D., Belhajjame, K., Cheney, J., Corsar, D., Garjjo, D., Soiland-Reyes, S., Zednik, S., Zhao, J.: PROV-O: The PROV Ontology. <https://www.w3.org/TR/prov-o/> (2013)
- [25] Bandrowski, A., Brinkman, R., Brochhausen, M., Brush, M.H., Bug, B., Chibucos, M.C., Clancy, K., Courtot, M., Dumontier, M., Haendel, M.A., *et al.*: The Ontology for Biomedical Investigations. *PLOS ONE* **11**(4), 0154556 (2016) <https://doi.org/10.1371/journal.pone.0154556>
- [26] Haller, A., Janowicz, K., Cox, S.J.D., Lefrançois, M., Taylor, K., Le Phuoc, D., Lieberman, J., García-Castro, R., Atkinson, R., Stadler, C.: Semantic Sensor Network Ontology. <https://www.w3.org/TR/vocab-ssn/> (2017)
- [27] Santangelo, B.E., Hegde, H., Caufield, J.H., Reese, J., Kliegr, T., Hunter, L.E., Lozupone, C.A., Mungall, C.J., Joachimiak, M.P.: KG-Microbe – building modular and scalable knowledge graphs for microbiome and microbial sciences. *GigaScience*, 077 (2026) <https://doi.org/10.1093/gigascience/giag077>
- [28] Ma, C., Liu, S., Won, S., Koslicki, D.: MetagenomicKG: a knowledge graph for

- metagenomic applications. *Bioinformatics* **42**(7), 421 (2026) <https://doi.org/10.1093/bioinformatics/btag421>
- [29] Open-Meteo: Open-Meteo: free weather API. <https://open-meteo.com/> (2024)
- [30] Dumontier, M., Baker, C.J., Baran, J., Callahan, A., Chepelev, L., Cruz-Toledo, J., Del Rio, N.R., Duck, G., Furlong, L.I., Keast, N., *et al.*: The SemanticScience Integrated Ontology (SIO) for biomedical research and knowledge discovery. *Journal of Biomedical Semantics* **5**(1), 14 (2014)
- [31] Degtyarenko, K., Matos, P., Ennis, M., Hastings, J., Zbinden, M., McNaught, A., Alcántara, R., Darsow, M., Guedj, M., Ashburner, M.: ChEBI: a database and ontology for chemical entities of biological interest. *Nucleic Acids Research* **36**(suppl.1), 453–462 (2008)
- [32] Buttigieg, P.L., Morrison, N., Smith, B., Mungall, C.J., Lewis, S.E.: The environment ontology: contextualising biological and biomedical entities. *Journal of Biomedical Semantics* **4**(1), 43 (2013)
- [33] Schoch, C.L., Ciuffo, S., Domrachev, M., Hotton, C.L., Kannan, S., Khovanskaya, O., Leipe, D., Mcveigh, R., O’Neill, K., Robbertse, B., *et al.*: NCBI Taxonomy: a comprehensive update on curation, resources and tools. *Database* **2020**, 062 (2020)
- [34] Tawfiq, R., Abdelhakim, M., Alarawi, M., Martinez Arbas, S., Alajel, S.M., Derouiche, A., Grünberg, R., Kulmanov, M., Niu, K., Lopez Velazquez, A., Wang, T., Rueping, M., Hoehndorf, R.: Landscape-scale bacterial biogeography across the Rub’ al-Khali reveals recurring spatial and soil-position structure. Companion ecology manuscript in preparation (2026)
- [35] Khan, T., Koning, K., Endresen, D., Chala, D., Kusch, E.: TwinEco: a unified framework for dynamic data-driven digital twins in ecology. *Ecological Informatics* **91**, 103407 (2025) <https://doi.org/10.1016/j.ecoinf.2025.103407>
- [36] Islam, S., Koivula, H., Andrew, C., Lopez Gordillo, J., Weiland, C., Schigel, D., Endresen, D., Arvanitidis, C., Chadwick, E., Soiland-Reyes, S.: FAIR digital twins for biodiversity: enabling data, model, and workflow integration. *npj Biodiversity* **5**(5) (2026) <https://doi.org/10.1038/s44185-025-00116-3>
- [37] Ovaskainen, O., Winter, S., Tikhonov, G., Lauha, P., Lehtiö, A., Nokelainen, O., Abrego, N., Aroluoma, A., Harrison, J.P., Heikkinen, M., *et al.*: A digital twin for real-time biodiversity forecasting with citizen science data. *Nature Ecology & Evolution* **10**, 481–495 (2026) <https://doi.org/10.1038/s41559-025-02966-3>
- [38] Klindworth, A., Pruesse, E., Schweer, T., Peplies, J., Quast, C., Horn, M., Glöckner, F.O.: Evaluation of general 16s ribosomal RNA gene PCR primers for classical and next-generation sequencing-based diversity studies. *Nucleic Acids*

- [39] Ewels, P.A., Peltzer, A., Fillinger, S., Patel, H., Alneberg, J., Wilm, A., Garcia, M.U., Di Tommaso, P., Nahnsen, S.: The nf-core framework for community-curated bioinformatics pipelines. *Nature Biotechnology* **38**(3), 276–278 (2020)
- [40] Straub, D., Blackwell, N., Langarica-Fuentes, A., Peltzer, A., Nahnsen, S., Kleindienst, S.: Interpretations of environmental microbial community studies are biased by the selected 16s rRNA (gene) amplicon sequencing pipeline. *Frontiers in Microbiology* **11**, 550420 (2020)
- [41] Quast, C., Pruesse, E., Yilmaz, P., Gerken, J., Schweer, T., Yarza, P., Peplies, J., Glöckner, F.O.: The SILVA ribosomal RNA gene database project: improved data processing and web-based tools. *Nucleic Acids Research* **41**(D1), 590–596 (2013) <https://doi.org/10.1093/nar/gks1219>
- [42] Bolyen, E., Rideout, J.R., Dillon, M.R., Bokulich, N.A., Abnet, C.C., Al-Ghalith, G.A., Alexander, H., Alm, E.J., Arumugam, M., Asnicar, F., *et al.*: Reproducible, interactive, scalable and extensible microbiome data science using QIIME 2. *Nature Biotechnology* **37**(8), 852–857 (2019)
- [43] Fierer, N., Leung, P.M., Lappan, R., Eisenhofer, R., Ricci, F., Holland, S.I., *et al.*: Guidelines for preventing and reporting contamination in low-biomass microbiome studies. *Nature Microbiology* **10**(7), 1570–1580 (2025) <https://doi.org/10.1038/s41564-025-02035-2>
- [44] Parks, D.H., Chuvochina, M., Rinke, C., Mussig, A.J., Chaumeil, P.-A., Hugenholtz, P.: GTDB: an ongoing census of bacterial and archaeal diversity through a phylogenetically consistent, rank normalized and complete genome-based taxonomy. *Nucleic Acids Research* **50**(D1), 199–207 (2022) <https://doi.org/10.1093/nar/gkab776>
- [45] Prud’hommeaux, E., Labra Gayo, J.E., Solbrig, H.: Shape Expressions: an RDF validation and transformation language. *Proceedings of the 10th International Conference on Semantic Systems*, 32–40 (2014)
- [46] Kazakov, Y., Krötzsch, M., Simančík, F.: The incredible ELK: From polynomial procedures to efficient reasoning with $\mathcal{EL}$ ontologies. *Journal of Automated Reasoning* **53**(1), 1–61 (2013) <https://doi.org/10.1007/s10817-013-9296-3>
- [47] Baader, F., Calvanese, D., McGuinness, D.L., Nardi, D., Patel-Schneider, P.F.: *The Description Logic Handbook: Theory, Implementation, and Applications*, 2nd edn. Cambridge University Press, Cambridge (2007). <https://doi.org/10.1017/CBO9780511711787>
- [48] Motik, B., Grau, B.C., Horrocks, I., Wu, Z., Fokoue, A., Lutz, C.: OWL 2 Web Ontology Language document overview. W3C Recommendation (2012). <https://www.w3.org/TR/owl2-overview/>

[//www.w3.org/TR/owl2-overview/](http://www.w3.org/TR/owl2-overview/)

- [49] Gkoutos, G.V., Schofield, P.N., Hoehndorf, R.: The Units Ontology: a tool for integrating units of measurement in science. *Database* **2012**, 033 (2012)
- [50] Gkoutos, G.V., Green, E.C., Mallon, A.-M., Hancock, J.M., Davidson, D.: The Phenotype And Trait Ontology (PATO): a standard for cross-species phenotype data integration. *Mammalian Genome* **16**, 866–879 (2005)
- [51] Horridge, M., Bechhofer, S.: The OWL API: a Java API for OWL ontologies. *Semantic Web* **2**(1), 11–21 (2011)
- [52] Grau, B.C., Horrocks, I., Kazakov, Y., Sattler, U.: A survey on ontology modularization. *Theoretical Computer Science* **391**(3), 198–233 (2008)
- [53] OpenLink Software: OpenLink Virtuoso Universal Server. <https://virtuoso.openlinksw.com/> (2024)
- [54] Horst, H.J.: Completeness, decidability and complexity of entailment for rdf schema and a semantic extension involving the owl vocabulary. *Journal of Web Semantics* **3**(2–3), 79–115 (2005) <https://doi.org/10.1016/j.websem.2005.06.001>